

Constraints on attractor models of inflation and reheating from *Planck*, BICEP/Keck, ACT DR6, and SPT-3G data

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We analyze the latest cosmic microwave background (CMB) constraints on the scalar spectral index n_s and tensor-to-scalar ratio r from *Planck* 2018, BICEP/Keck 2018, the Atacama Cosmology Telescope Data Release 6 (ACT DR6), and the South Pole Telescope (SPT-3G) data, focusing on their implications for attractor models of inflation. We compare systematically observational bounds with theoretical predictions for both E-model (α -Starobinsky) and T-model potentials. The observational constraints accommodate E-models with $\alpha \lesssim 25$, with the canonical Starobinsky model ($\alpha = 1$) predicting $n_s = 0.958\text{--}0.963$ for reheating temperatures between $100 - 10^{10}$ GeV, in good agreement with *Planck* 2018 data and within the 95% CL region determined by the *Planck*-ACT-SPT combination, but below the 95% confidence region of the *Planck*-ACT-DESI combination. Higher reheating temperatures from near-instantaneous reheating improve the compatibility. T-models predict slightly lower n_s values (0.956–0.961), in some tension with *Planck* 2018 data, and we find an upper limit of $\alpha \lesssim 11$ in these models. We extend our analysis to generalized α -attractors with monomial potentials $V(\varphi) \propto \varphi^k$ near the minimum, demonstrating that models with $k \geq 6$ naturally predict $n_s \simeq 0.965\text{--}0.968$ for typical number of e -folds, in better agreement with the ACT DR6 data. We also consider deformed E- and T-models, which allow significantly higher values of n_s for low values of $\alpha \simeq 1$.

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I. INTRODUCTION

Inflation, a period of accelerated expansion in the early Universe, provides a compelling explanation for the origin

of primordial density fluctuations and the near-flatness and homogeneity of the observable Universe [1]. These two predictions became testable when observations of the total mass-energy density relative to the critical density corresponding to a flat Universe, Ω_{tot} , and the spectral tilt, n_s , of the scalar anisotropy spectrum, were determined first by WMAP [2] and then with significantly higher precision by *Planck* [3]. While most models of inflation predict that Ω_{tot} should be indistinguishable from unity, measurements of the spectral tilt and upper limits on the ratio of tensor and scalar perturbation amplitudes, r , have become important discriminators between models [3–5]. Many models of inflation could be excluded by their combination.

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Until recently, the CMB data appeared to favor the original inflationary model of Starobinsky [6] as being largely consistent with the determination of n_s and the upper limit on r . The Starobinsky model, initially formulated as a means to avoid the initial big bang singularity, can be expressed as a modification of the Einstein-Hilbert action for gravity, namely

$$\mathcal{S} = \frac{M_P^2}{2} \int d^4x \sqrt{-g} \left(-R + \frac{R^2}{6M^2} \right), \quad (1)$$

where R is the Ricci scalar, M is the inflationary mass scale, and $M_P = (8\pi G)^{-1/2} \simeq 2.435 \times 10^{18}$ GeV is the reduced Planck mass.

This theory may be rewritten in the Einstein frame as Einstein-Hilbert gravity with a supplementary canonical scalar field, the inflaton, φ , (sometimes referred to as the scalaron in this context) [7–9] with a scalar potential given by

$$V = \frac{3}{4} M^2 M_P^2 \left(1 - e^{-\sqrt{\frac{2}{3}} \frac{\varphi}{M_P}} \right)^2. \quad (2)$$

The inflationary mass scale M (also the inflaton mass for this potential) is determined by the amplitude of the scalar fluctuation spectrum,

$$A_s = \frac{3M^2}{8\pi^2 M_P^2} \sinh^4 \left(\frac{\varphi_*}{\sqrt{6} M_P} \right), \quad (3)$$

where $\varphi_* \simeq 5.35 M_P$ is the value of the inflaton field prior to the final ~ 55 e -folds of inflation (the determination of φ_* will be discussed in greater detail below) and $A_s \simeq 2.1 \times 10^{-9}$ [3], in which case $M = 1.25 \times 10^{-5} M_P \simeq 3 \times 10^{13}$ GeV. The inflationary slow-roll parameters determined from φ_* (also discussed below) fix the scalar tilt, $n_s = 0.965$ and a tensor-to-scalar ratio $r = 0.0035$.

The *Planck* satellite experiment [3] in combination with gravitational lensing measured the scalar tilt as

$$n_s = 0.9649 \pm 0.0042 \text{ (68\% CL)}, \quad (4)$$

and *Planck* in combination with observations by BICEP/Keck [4] provided an upper limit on r

$$r < 0.036, \quad (5)$$

both in excellent agreement with the predictions of the Starobinsky model.¹

Recently, two ground-based CMB experiments have released new results that impact the value of n_s . Both the Atacama Cosmology Telescope (ACT) [10,11] and the

South Pole Telescope (SPT) [12] have released high resolution data at small angular scales (large multipoles). These alone are neither very different nor more accurate than the *Planck* result for n_s . However, when combined with *Planck* results that include large angular scales, the derived values of n_s shift (particularly in the combination of *Planck* and ACT data when DESI DR1 data [13,14] and DR2 data [15,16]) with smaller uncertainties. In the following we will discuss the implications of the BICEP/Keck 2018 results which incorporate *Planck* and WMAP [4] with the ACT and SPT results.

Among the diverse landscape of inflationary theories, no-scale supergravity [17,18] has emerged as a particularly compelling framework, capable of mimicking the predictions for n_s and r of the Starobinsky model [19], and also able to accommodate generalizations such as attractor models [20–23]. These models naturally produce a flat potential plateau at large field values that generates inflationary dynamics consistent with *Planck* measurements.

Two specific types of attractor potentials have been widely studied [21–26]: α -Starobinsky models, which are also referred to as E-models in the literature [25]

$$V = \frac{3}{4} \lambda M_P^4 \left(1 - e^{-\sqrt{\frac{2}{3}} \frac{\varphi}{M_P}} \right)^2, \quad (6)$$

and T-models [26]

$$V = \frac{3}{4} \lambda M_P^4 \tanh^2 \left(\frac{\varphi}{\sqrt{6} \alpha M_P} \right), \quad (7)$$

where λ is determined by the CMB normalization and corresponds to M^2/M_P^2 in the Starobinsky model, which is a special case of Eq. (6) with $\alpha = 1$.² Generalized versions of these models are discussed in Sec. V.

In response to the more recent CMB results, a wave of studies has reexamined the wider inflationary landscape. For example, Ref. [27] demonstrated that a simple generalization of chaotic inflation with nonminimal coupling to gravity provides a good match to ACT results with $r \simeq 10^{-2}$, while comprehensive analyses of slow-roll models reveal that the combined ACT DR6, *Planck*, DESI, and BICEP/Keck datasets place increased pressure on conventional inflationary scenarios [28–46]. The ACT data exclude standard Starobinsky and Higgs inflation models at the 2σ level, motivating investigations of reheating effects [47–51], curvature corrections to Starobinsky inflation that can reconcile predictions with ACT observations [52–59], and polynomial potential inflation models that remain viable [60–62]. Studies of some attractor models

¹We note that a more stringent bound of $r < 0.032$ was obtained in [5].

²We adopt a normalization convention in which the normalization scales of both inflaton potentials coincide, as detailed in Eq. (38). This choice leaves the predicted spectral index n_s and tensor-to-scalar ratio r unchanged.

show that radiative corrections may shift predictions into the ACT-favored parameter space [63–68] and smooth hybrid inflation models that naturally predict spectral indices around 0.97 [69], collectively demonstrating how precision cosmological measurements continue to refine theoretical understanding of the early universe and challenge established paradigms in inflationary cosmology.

Overall, the consensus emerging from recent works is that *modified* plateau potentials, whether via higher-order exponents, nonminimal couplings, or modest multifield extensions, remain fully compatible with the ACT DR6 + *Planck* + BK18 data, while simple chaotic monomials and the original R^2 model are increasingly challenged. The sensitivity of the planned LiteBIRD [70] mission to $r \sim \mathcal{O}(10^{-3})$ will provide a decisive test for this refined class of inflationary scenarios.

In this paper, we explore the impact of the latest BICEP/Keck/WMAP/*Planck*/ACT/SPT constraints in the (n_s, r) plane on the E-model and T-model inflationary attractor models. We find that the increase in n_s from the *Planck*/ACT/DESI DR1 (P-ACT-LB) combination creates significant tension with standard α -attractor predictions, whereas the *Planck*/ACT/SPT (CMB-SPA) results remain consistent with the conventional expectations. We examine systematically the allowed values of the α parameter, paying particular attention to the reheating temperature that, together with the reheating equation of state w_{reh} , fixes the e -folding number N_* . The allowed range of N_* is determined by the minimum reheating temperature, which we take to be the electroweak scale, ~ 100 GeV, and the maximum reheating temperature, which we take as $\sim 10^{10}$ GeV, motivated by supergravity models with a 100 GeV gravitino and supersymmetric dark matter [71,72], as discussed in more detail later. In supersymmetric realizations where the gravitino bound enforces $T_{\text{RH}} \lesssim 10^{10}$ GeV, the corresponding reduction in N_* lowers the predicted n_s , thereby strengthening the tension between the canonical no-scale ($\kappa = 1, k = 2$) supergravity attractors and ACT-favored constraints (quantified in Sec. VII). We also consider an extended range of reheating temperatures ranging from the big bang nucleosynthesis (BBN) limit, 4 MeV [73] to instantaneous reheating, $\sim 2 \times 10^{15}$ GeV. We extend our analysis to generalized attractor models with monomial potentials $V(\varphi) \propto \varphi^k$ near the minimum, demonstrating that models with $k \geq 6$ can accommodate more easily the ACT preference for larger n_s . It has also been shown that slightly deformed potentials [32], that occur quite naturally when the inflation model is embedded in UV completion such as an SU(5) or SO(10) GUT theory [74], may have a significant impact on the CMB observables, as we discuss in more detail below.

The remainder of this paper is organized as follows. In Sec. II we introduce and review no-scale models of inflation. In Sec. III, we compile and analyze the current observational constraints on (n_s, r) from various CMB experiments.

In Sec. IV, we present the inflationary dynamics and derive CMB observables. In Sec. V, we analyze generalized α -attractors with higher-order minima as well as deformed models. In Sec. VI, we discuss the reheating constraints we use to limit N_* . We present the results in Sec. VII and conclude in Sec. VIII.

II. REVIEW OF NO-SCALE SUPERGRAVITY MODELS OF INFLATION

There is a deep correspondence between the $R + R^2$ theory and no-scale supergravity [19,21,75–79], and several well-motivated scenarios arising naturally in supergravity and string theory appear within the α -attractor framework. The simplest no-scale supergravity [17,18] models are characterized by a Kähler potential of the form

$$K = -3\alpha M_P^2 \ln \left(\frac{T + \bar{T}}{M_P} - \frac{|\phi|^2}{3M_P^2} \right), \quad (8)$$

where $\alpha = 1$ in the simplest model [19], but $\alpha \neq 1$ is possible in generalized models [21]. Here, T is the volume modulus and ϕ is a matterlike field. Exploiting the underlying $\text{SU}(2, 1)/\text{SU}(2) \times \text{U}(1)$ symmetry, the Starobinsky potential can be derived from many different superpotentials [19,21,80]. For example, when $\alpha = 1$

$$W = M \left(\frac{1}{2} \phi^2 - \frac{1}{3\sqrt{3}M_P} \phi^3 \right), \quad (9)$$

leads to the Starobinsky potential [19] with the canonically normalized inflaton related to ϕ through

$$\phi = \sqrt{3}M_P \tanh \left(\frac{\varphi}{\sqrt{6}M_P} \right). \quad (10)$$

Alternatively (also when $\alpha = 1$),

$$W = \sqrt{3}MM_P\phi \left(\frac{T}{M_P} - \frac{1}{2} \right) \quad (11)$$

leads to the Starobinsky potential [21,81] when the canonically normalized inflaton is related to T through

$$T = \frac{M_P}{2} e^{\sqrt{\frac{2}{3}} \frac{\varphi}{M_P}}. \quad (12)$$

The α -attractor variations of the Starobinsky model can be derived in no-scale supergravity as discussed in more detail in Appendix A. Such variations arise naturally in compactified string models [82], where the volume modulus T factorizes as a product of three independent moduli T_i . Models where inflation is driven by one (two) of these moduli correspond to $\alpha = 1/3$ ($2/3$) [21], while larger values of α can arise from complex structure moduli [20]. The parameter α that determines the (constant) curvature of the Kähler manifold plays a crucial role in determining the inflationary predictions: larger values of α reduce the

flatness of the potential plateau at the horizon-crossing field value φ_* , thereby modifying the observables n_s and r . Generalizing the coefficient of the log in the Kähler potential to $\alpha \neq 1$ modifies the prediction for r by a factor α , as first pointed out in [21] and subsequently in [23]. Remarkably, as demonstrated in [21–24,78,83], the predictions of all (α)-attractor models converge to universal values in the large- N_* limit

$$n_s \simeq 1 - \frac{2}{N_*}, \quad r \simeq \frac{12\alpha}{N_*^2}, \quad (13)$$

where the relation between φ_* and N_* is discussed in greater detail in Sec. IV below.

This universality arises because the α -Starobinsky (E-model) and T-model potentials have identical leading-order behaviors in their small-field expansions.³ For the canonical Starobinsky model with $\alpha = 1$, the predicted values $n_s \simeq 0.961$ – 0.968 and $r \simeq 0.004$ – 0.003 (for $N_* = 50$ – 60 e -folds) were long considered a triumph of the theory, aligning perfectly with pre-ACT CMB measurements. However, these predictions now fall significantly below the P-ACT-LB2 central value of $n_s = 0.9752$, potentially signaling a departure from the previously established concordance.

A comprehensive analysis of α -attractor models using BICEP/Keck 2018 data was performed in [84], which found that both α -Starobinsky and T-models remained comfortably within the observational bounds available at that time. The study demonstrated that α -Starobinsky models with $\alpha \in (0.67, 12)$ and T-models with $\alpha \in (1.3, 5.1)$ fell within the 68% confidence region, while considering gravitino and dark matter constraints from supergravity. Prior to ACT DR6, these values sat comfortably within the confidence regions of all major CMB experiments, establishing Starobinsky inflation as a benchmark model. For a related study see Ref. [85].

III. OBSERVATIONAL CONSTRAINTS

Multiple CMB experiments have provided increasingly precise measurements of the scalar spectral index n_s and stringent upper limits on the tensor-to-scalar ratio r . Table I summarizes the current constraints from key datasets.

A. Planck 2018

The *Planck* satellite 2018 data release [3] remains the baseline for CMB parameter constraints. The analysis of temperature and polarization data yields $n_s = 0.9649 \pm 0.0044$ (68% CL) from the TT, TE, EE + low

³The potentials (6) and (7) agree at first order in $\exp(-\sqrt{2/(3\alpha)}\varphi/M_P)$ but differ at higher orders, yielding model-dependent corrections when $\varphi \sim \sqrt{\alpha}M_P$. These corrections become negligible for $N_* \gg 1$.

TABLE I. Recent CMB constraints on the scalar spectral index n_s (68% CL) and the tensor-to-scalar ratio r (95% CL). Unless otherwise noted, values are quoted at a pivot scale $k_* = 0.05 \text{ Mpc}^{-1}$.

Dataset	n_s	r
<i>Planck</i> 2018 (TT, TE, EE + lowE) ^a	0.9649 ± 0.0044	$< 0.11^b$
<i>Planck</i> 2018 + lensing ^a	0.9649 ± 0.0042	$< 0.10^b$
<i>Planck</i> 2018 + lensing + BAO ^a	0.9665 ± 0.0038	$< 0.11^b$
<i>Planck</i> 2018 + BK15 ^a + lensing + BAO ^a	0.9668 ± 0.0037	$< 0.058^b$
<i>Planck</i> 2018 + BK18 ^c	...	< 0.036
ACT DR6 ^d	0.9666 ± 0.0077	...
ACT DR6 + WMAP ^d	0.9660 ± 0.0046	...
<i>Planck</i> + ACT DR6 ^d	0.9709 ± 0.0038	...
<i>Planck</i> + ACT DR6 + lensing + BAO ^d	0.9752 ± 0.0030	...
<i>Planck</i> + ACT DR6 + BK18 ^e	...	< 0.038
SPT-3G (2019-2020) ^f	...	< 0.25
SPT-3G + <i>Planck</i> ^g	0.9647 ± 0.0037	...
SPT-3G D1 ^h	0.951 ± 0.011	...
SPT-3G D1 + ACT ^h	0.9671 ± 0.0058	...
SPT-3G D1 + <i>Planck</i> ^h	0.9636 ± 0.0035	...
CMB-SPA ^h	0.9684 ± 0.0030	...

^a[3];

^bAt a pivot scale $k_* = 0.002 \text{ Mpc}^{-1}$;

^c[4];

^d[10];

^e[11];

^f[86];

^g[87];

^h[12].

E + likelihood, with negligible change when including CMB lensing as given in Eq. (4). This is an 8σ detection of a red-tilted spectrum ($n_s < 1$) and definitively rules out exact scale invariance. For tensor modes, *Planck* alone constrains $r_{0.002} < 0.11$ (95% CL). This limit was strengthened by the BICEP/Keck 2015 result of $r < 0.062$. When combined with BICEP/Keck 2015 data, the *Planck* result tightens to $r_{0.002} < 0.058$ (95% CL) with a slight shift in the spectral index to $n_s = 0.9668 \pm 0.0037$ [3]. These constraints exclude simple polynomial potentials such as $V(\phi) \propto \phi^4$ at more than 3σ and favor models with plateaulike potentials, particularly those with $V''(\phi) < 0$ during inflation.

B. BICEP/Keck 2018

The BICEP/Keck Collaboration targets degree-scale B-mode polarization from a dedicated South Pole observatory, providing the most stringent constraints on inflationary gravitational waves. The progression of limits illustrates the rapid experimental progress: BICEP2+Keck+*Planck* (data up to 2014) constrained $r_{0.05} < 0.09$ (95% CL) [88], while BICEP/Keck data up to 2015 (BK15) combined with *Planck* constrained $r_{0.05} < 0.062$ (95% CL) [89]. The latest analysis

incorporating BICEP3 and Keck Array observations up to 2018 (BK18) yields the current best limit: $r_{0.05} < 0.036$ (95% CL) [4],⁴ and a likelihood analysis yielded $r_{0.05} = 0.014^{+0.010}_{-0.011}$. These constraints assume the inflationary consistency relation $n_t = -r/8$. The constraint corresponds to an uncertainty $\sigma(r_{0.05}) = 0.009$, approaching the sensitivity needed to detect $r \sim 0.01$ models. While BICEP/Keck observations do not significantly constrain n_s (which is determined by large-scale temperature and E -mode data), they provide crucial complementarity to *Planck*.

C. ACT DR6

The ACT Data Release 6 produced high-resolution maps of the CMB temperature and polarization measurements over 40% of the sky, complementing *Planck*'s full-sky coverage with superior small-scale sensitivity [10,11]. ACT's arcmin resolution probes small-scale multipoles complementary to *Planck*. These data have been combined with *Planck* and WMAP in order to extend down to low multipoles and are in good agreement with both *Planck* and WMAP larger-scale data. For example, the combination of WMAP and ACT (W-ACT) yielded⁵ $n_s = 0.9660 \pm 0.0046$, whereas *Planck* without lensing data give $n_s = 0.9649 \pm 0.0044$. However, because the large-scale data result in a positive correlation between n_s and the baryon density $\Omega_b h^2$, in contrast to them being anticorrelated in the ACT data, and since the W-ACT determination of the baryon density is slightly higher than that of *Planck* alone ($\Omega_b h^2 = 0.02263 \pm 0.00012$ vs 0.02237 ± 0.00015), the combination of *Planck* and ACT (P-ACT) tends to increase the fit value of $n_s = 0.9709 \pm 0.0038$ and P-ACT with lensing gives 0.9713 ± 0.0037 , about $1.5 - 2\sigma$ higher than the *Planck* result.

ACT has also combined their data with DESI DR1 data [13,14] and subsequently DR2 data [15,16]. These combinations come with potential additional uncertainties due to unknown systematics in the supernovae data (see, e.g., [91]). This combination has yielded new CMB constraints that support inflation broadly while further challenging specific models. Most notably, these data imply a higher baryon density of $\Omega_b h^2 = 0.02256 \pm 0.00011$ for the *Planck*-ACT-DESI DR1 (P-ACT-LB) combination of data and $\Omega_b h^2 = 0.02258 \pm 0.00010$ using DESI DR2 (P-ACT-LB2). This shifts the scalar spectral index n_s significantly upward from previous measurements. The combination of *Planck* + lensing, ACT, and DESI BAO (P-ACT-LB) pushes the tilt to $n_s = 0.9743 \pm 0.0034$ and even higher using DR2 data (P-ACT-LB2) [10]

⁴A subsequent analysis in [5] used BB autocorrelation data from [90] and obtained $r_{0.05} < 0.032$, allowing for a free reionization optical depth.

⁵Unless specifically noted all uncertainties are given as 68% CL.

$$n_s = 0.9752 \pm 0.0030 \quad (68\% \text{ CL}). \quad (14)$$

The P-ACT-LB2 results lie approximately 2σ above the *Planck* 2018 constraint, representing a significant shift in their preferred value of n_s . Some caution is warranted, however, as the shift may be reflective of underlying tension between BAO/CMB analyses [92].

For tensor constraints, the ACT DR6 measurements, which favor $n_s \simeq 0.975$, lack direct sensitivity to degree-scale B -modes and rely on the BICEP/Keck likelihood. The combined *Planck*+ACT+BK18 analysis maintains $r < 0.038$ (95% CL) [10], with the modest relaxation from BK18 alone reflecting parameter degeneracies in the expanded dataset. This combination of n_s and r places the canonical Starobinsky model ($\alpha = 1$) outside the 95% confidence region, indicating a $\sim 2\sigma$ tension with its theoretical predictions [11]. At $r = 0.004$, the P-ACT-LB 95% CL range on n_s is $0.967 < n_s < 0.982$. As noted above, this result merits careful consideration, particularly given the variation among more recent CMB analyses.

D. SPT-3G

The third-generation South Pole Telescope camera (SPT-3G) represents a major upgrade in polarization sensitivity and sky coverage. Initial results from the 2019-2020 observing seasons demonstrate the instrument's capabilities. SPT-3G measures the B -mode power spectrum over $30 < \ell < 500$, achieving $r < 0.25$ (95% CL) [86] from these data alone. While this constraint is weaker than BICEP/Keck results, it represents only 5% of the planned dataset and serves primarily as a systematic check and foreground characterization.

While SPT alone does not determine n_s at high precision (the uncertainty is 0.011), the combination with *Planck* yields competitive cosmological constraints: $n_s = 0.9647 \pm 0.0037$ (68% CL) from SPT+*Planck* [87], consistent with the original *Planck* value and in tension with ACT DR6. The latest SPT-3G analysis incorporating data through 2020 confirms a similar result, with $n_s \simeq 0.9636 \pm 0.0035$ in combination with *Planck* [12]. This is also significantly lower than the P-ACT value.

The combined datasets of SPT+*Planck*+ACT (denoted CMB-SPA) yield [12] the following spectral tilt:

$$n_s = 0.9684 \pm 0.0030 \quad (68\% \text{ CL}), \quad (15)$$

which is consistent with the *Planck* result (at 1σ). Adding DESI BAO results in $n_s = 0.9728 \pm 0.0027$, somewhat lower than the P-ACT-LB result. While further data (e.g., extended DESI analyses and SPT-3G observations) will help clarify these differences, the current tension between the ACT-shifted constraints and standard Starobinsky model predictions requires theoretical examination.

However, a new 2.8σ discrepancy has emerged in the Λ CDM between the combined CMB experiments and DESI BAO measurements, suggesting potential tensions in our understanding of late-time cosmology. This discrepancy between contemporary ground-based experiments and BAO data highlights the importance of systematic cross-checks and motivates exploration of extended cosmological models. Looking ahead, SPT-3G aims for a tensor sensitivity $\sigma(r) \sim 0.003$ through delensing techniques and five years of integrated observations, potentially accessing $r \sim 0.01$ inflationary models. The expanded SPT-3G survey will eventually cover 25% of the sky, dramatically improving constraints and enabling more stringent tests of cosmological models.

In what follows, we compare the predictions of models of inflation with the three CMB results that are highlighted in Eqs. (4), (14), and (15). Note that only the P-ACT-LB results make use of BAO data.

IV. INFLATIONARY DYNAMICS, CMB OBSERVABLES, AND REHEATING

The dynamics of the inflaton field φ is governed by the action

$$S = \int d^4x \sqrt{-g} \left[-\frac{M_P^2}{2} R + \frac{1}{2} g^{\mu\nu} \partial_\mu \varphi \partial_\nu \varphi - V(\varphi) \right], \quad (16)$$

where $V(\varphi)$ is the inflaton potential given by Eq. (6) for the α -Starobinsky (E-model) or Eq. (7) for the T-model. In the slow-roll approximation, the inflationary dynamics is characterized by the slow-roll parameters

$$\varepsilon_V \equiv \frac{M_P^2}{2} \left(\frac{V'}{V} \right)^2, \quad \eta_V \equiv M_P^2 \frac{V''}{V}, \quad (17)$$

where primes denote derivatives with respect to φ . During slow-roll inflation, these parameters satisfy $\varepsilon_V, |\eta_V| \ll 1$. The number of e -folds from horizon exit of the pivot scale to the end of inflation is given by

$$\begin{aligned} N_* &= \int_{t_*}^{t_{\text{end}}} H dt \simeq \frac{1}{M_P^2} \int_{\varphi_*}^{\varphi_{\text{end}}} \frac{V(\varphi)}{V'(\varphi)} d\varphi \\ &\simeq \int_{\varphi_{\text{end}}}^{\varphi_*} \frac{1}{\sqrt{2\varepsilon_V} M_P} d\varphi, \end{aligned} \quad (18)$$

where φ_* denotes the field value when the pivot scale k_* exits the horizon. The *Planck* Collaboration adopted $k_* = 0.05 \text{ Mpc}^{-1}$, and we also adopt this except if stated otherwise. The end of inflation is defined by the condition $\ddot{a} = 0$, where $a(t)$ is the scale factor corresponding to $\varepsilon_H(\varphi_{\text{end}}) = 1$, and $\varepsilon_H \equiv (M_P^2/2)(H'/H)^2$. In terms of ε_V , inflation ends when [93]

$$\varepsilon_V \simeq (1 + \sqrt{1 - \eta_V/2})^2. \quad (19)$$

The primary CMB observables are expressed in terms of the slow-roll parameters evaluated at horizon exit

$$n_s \simeq 1 - 6\varepsilon_{V*} + 2\eta_{V*}, \quad (20)$$

$$r \simeq 16\varepsilon_{V*}, \quad (21)$$

$$A_s = \frac{V_*}{24\pi^2 \varepsilon_{V*} M_P^4}, \quad (22)$$

where the asterisk indicates quantities evaluated at φ_* . The amplitude of scalar perturbations is observationally constrained to $\ln(10^{10} A_s) = 3.044 \pm 0.014$ at 68% CL from *Planck* 2018 TT, TE, EE + lowE + lensing data [3], corresponding to $A_s \simeq 2.10 \times 10^{-9}$. For the Starobinsky potential Eq. (22) is equivalent to Eq. (3).

For the attractor models described by Eqs. (6) and (7), the predictions in the large- N_* limit take the particularly simple form given by Eq. (13) [21]. These expressions are valid for $\alpha \lesssim \mathcal{O}(1)$ and $N_* \gg 1$. For the Starobinsky limit ($\alpha = 1$), more precise analytical expressions including higher-order corrections can be found in Ref. [72]. We note that the prediction for n_s is independent of α to leading order, making it a robust prediction of this class of models.

Using the e -fold expression (18), we can derive the inflaton field value at horizon crossing for the pivot scale k_* . For the case $\alpha = 1$, the field values are given by [93]

$$\begin{aligned} \frac{\varphi_*}{M_P} &= \sqrt{\frac{3}{2}} \left[1 + \frac{3}{4N_* - 3} \right] \\ &\times \ln \left(\frac{4N_*}{3} + e^{\sqrt{\frac{2}{3}} \frac{\varphi_{\text{end}}}{M_P}} - \sqrt{\frac{2}{3}} \frac{\varphi_{\text{end}}}{M_P} \right), \end{aligned} \quad (23)$$

$$\begin{aligned} \frac{\varphi_{\text{end}}}{M_P} &= \sqrt{\frac{3}{2}} \ln \left[\frac{2}{11} (4 + 3\sqrt{3}) \right] \simeq 0.63, \\ [\alpha\text{-Starobinsky (E-model)}], \end{aligned} \quad (24)$$

$$\frac{\varphi_*}{M_P} = \sqrt{\frac{3}{2}} \cosh^{-1} \left(\frac{4N_*}{3} + \cosh \left(\sqrt{\frac{2}{3}} \frac{\varphi_{\text{end}}}{M_P} \right) \right), \quad (25)$$

$$\begin{aligned} \frac{\varphi_{\text{end}}}{M_P} &= \sqrt{\frac{3}{2}} \ln \left[\frac{1}{11} (14 + 5\sqrt{3}) \right] \simeq 0.89, \\ (\text{T-model}). \end{aligned} \quad (26)$$

Full analytical expressions including the dependence on α are provided in Appendix B, Eqs. (B1)–(B6).

We next calculate the number of e -folds N_* between horizon exit and the end of inflation, accounting for the postinflationary evolution. Assuming no additional entropy production between the end of reheating and horizon reentry, we have [94,95]

$$\begin{aligned}
N_* = & \ln \left[\frac{1}{\sqrt{3}} \left(\frac{\pi^2}{30} \right)^{1/4} \left(\frac{43}{11} \right)^{1/3} \frac{T_0}{H_0} \right] - \ln \left(\frac{k_*}{a_0 H_0} \right) \\
& - \frac{1}{12} \ln g_{\text{RH}} + \frac{1}{4} \ln \left(\frac{V_*^2}{M_P^4 \rho_{\text{end}}} \right) \\
& + \frac{1 - 3w_{\text{int}}}{12(1 + w_{\text{int}})} \ln \left(\frac{\rho_{\text{rad}}}{\rho_{\text{end}}} \right), \quad (27)
\end{aligned}$$

where $H_0 = 67.36 \text{ km s}^{-1} \text{ Mpc}^{-1}$ [3] and $T_0 = 2.7255 \text{ K}$ [96] are the present-day Hubble parameter determined by *Planck* and the CMB temperature, respectively. Here, $\rho_{\text{end}} = 3M_P^2 H_{\text{end}}^2$ is the energy density at the end of inflation, and ρ_{rad} is the energy density in radiation in the radiation-dominated era when $w = p/\rho = 1/3$. Up to a subpercent correction in N_* , this energy density can be associated with the energy density at the end of reheating, ρ_{RH} , defined when energy densities in radiation and the inflaton are equal, $\rho_\phi = \rho_{\text{R}} \equiv \rho_{\text{RH}}$. In turn, this can be related to the reheating temperature T_{RH} via the thermodynamic relation $\rho_{\text{RH}} = (g_{\text{RH}} \pi^2/30) T_{\text{RH}}^4$, where g_{RH} denotes the number of effective relativistic degrees of freedom at reheating, which is $g_{\text{reh}} = 915/4$ in the minimal supersymmetric extension of the Standard Model (MSSM), and $a_0 = 1$ is the scale factor at the present day. The equation of state parameter during reheating is characterized by its average

$$w_{\text{int}} \equiv \frac{1}{N_{\text{rad}} - N_{\text{end}}} \int_{N_{\text{end}}}^{N_{\text{rad}}} w(n) dn. \quad (28)$$

For a detailed discussion on the computation of N_* , see Appendix B. For the *Planck* pivot scale $k_* = 0.05 \text{ Mpc}^{-1}$, the first two terms in Eq. (27) yield

$$N_* = 61.49 - \frac{1}{12} \ln g_{\text{RH}} + \Delta N_{\text{inf}} + \Delta N_{\text{rad}}, \quad (29)$$

where ΔN_{inf} and ΔN_{rad} represent the model-dependent contributions from inflation and reheating, respectively.

We consider three benchmark reheating scenarios

- (i) MSSM reheating: Taking $g_{\text{RH}} = 915/4$ corresponding to the MSSM degrees of freedom, we obtain

$$N_*^{\text{MSSM}} = 61.04 + \dots \quad (30)$$

- (ii) Electroweak-scale reheating: For $T_{\text{reh}} = 100 \text{ GeV}$ with SM degrees of freedom ($g_{\text{RH}} = 427/4$), we find

$$N_*^{\text{EW}} = 61.10 + \dots \quad (31)$$

- (iii) BBN compatible reheating: The minimum temperature consistent with BBN is $T_{\text{reh}} \gtrsim \mathcal{O}(1) \text{ MeV}$. Taking $T_{\text{BBN}} = 4 \text{ MeV}$ with $g_{\text{RH}} = 10.75$, we obtain

$$N_*^{\text{BBN}} = 61.29 + \dots \quad (32)$$

As we shall see, typical values of $N_* \simeq 45\text{--}55$ encompasses these scenarios for typical inflationary potentials and reheating dynamics with $-1/3 \leq w_{\text{int}} \leq 1$.

To determine N_* numerically for different reheating scenarios, we solve the coupled system of equations governing the postinflationary dynamics. The evolution of the inflaton and radiation energy densities during reheating is described by

$$\dot{\rho}_\phi + 3H(1 + w_\phi)\rho_\phi = -\Gamma_\phi(1 + w_\phi)\rho_\phi, \quad (33)$$

$$\dot{\rho}_r + 4H\rho_r = \Gamma_\phi(1 + w_\phi)\rho_\phi, \quad (34)$$

$$\rho_\phi + \rho_r = 3M_P^2 H^2, \quad (35)$$

where Γ_ϕ is the inflaton decay rate, and w_ϕ is the inflaton equation of state parameter. The time-averaged equation of state during reheating evolves according to

$$\frac{d}{dt}(Nw_{\text{int}}) = Hw, \quad (36)$$

where $w = (\rho_\phi w_\phi + \rho_r/3)/(\rho_\phi + \rho_r)$ is the instantaneous total equation of state.

We assume that reheating occurs through the perturbative decay of the inflaton to either scalars or fermions. For potentials with an approximate quadratic minimum [as in Eqs. (6) and (7)] the reheating temperature is determined by the decay rate, Γ_ϕ ,

$$\frac{g_{\text{RH}} \pi^2}{30} T_{\text{RH}}^4 = \frac{12}{25} \Gamma_\phi^2 M_P^2. \quad (37)$$

For the generalized models discussed in the next section, the corresponding expressions for T_{RH} in terms of the decay rate have been derived in Ref. [97]. We allow the reheating temperature to take values as low as 4 MeV to ensure standard BBN [73] and in principle as high as the perturbative limit at $T_{\text{RH}} \sim 10^{15} \text{ GeV}$. A more restrictive range is $100 \lesssim T_{\text{RH}}/\text{GeV} \lesssim 10^{10}$, where the lower limit would allow for leptogenesis such that sphaleron interactions reach equilibrium [98] and the upper limit could be imposed in supersymmetric models to avoid the overproduction of a 100 GeV gravitino and supersymmetric dark matter [71,72].

Given the value of ϕ_* , determined from Eqs. (18) and (27), we can combine these field values with the amplitude constraint from Eq. (22) to obtain the normalization of the inflaton potential, which to a good approximation is given by

$$\lambda \simeq \frac{24a\pi^2 A_s}{N_*^2}. \quad (38)$$

Here λ is the dimensionless potential normalization parameter (the overall height) appearing in Eqs. (39)–(40) via

$V_0 = \frac{3}{4}\lambda M_P^4$ and $A_s \simeq 2.1 \times 10^{-9}$. This demonstrates that the energy scale of inflation is proportional to $\sqrt{\alpha}$, with larger α corresponding to higher inflationary energy scales. For the canonical Starobinsky model ($\alpha = 1$, $N_* = 55$), this yields $\lambda \simeq 1.6 \times 10^{-10}$, corresponding to an inflationary Hubble scale $H_* \simeq 1.5 \times 10^{13}$ GeV.

V. INFLATIONARY ATTRACTORS

A. Generalized attractor models

The predictions of the canonical E- and T-models (6), (7) which yield $n_s \simeq 0.957\text{--}0.963$ for reheating temperatures between $100\text{--}10^{10}$ GeV, as we demonstrate in Sec. VII, are significantly below the value of $n_s \simeq 0.975$ preferred by the ACT DR6 data, and motivates consideration of generalized attractor potentials [24,26,72]

$$V = \frac{3}{4}\lambda M_P^4 \left(1 - e^{-\sqrt{\frac{2}{3\alpha}}\frac{\varphi}{M_P}}\right)^k, \quad \text{[Generalized } \alpha\text{-Starobinsky (E-model)]} \quad (39)$$

$$V = \frac{3}{4}\lambda M_P^4 \tanh^k \left(\frac{\varphi}{\sqrt{6\alpha}M_P} \right), \quad \text{(Generalized T-model)} \quad (40)$$

where $k \geq 2$ is an even integer. These potentials retain the attractor behavior at large field values while modifying the approach to the minimum, with $V(\varphi) \propto \varphi^k$ for small φ . As a consequence, the inflaton mass near the minimum is given by

$$m_\varphi^{(E)} = \frac{\sqrt{3}}{2} \sqrt{\left(\frac{2}{3}\right)^{\frac{k}{2}} k(k-1) \alpha^{-\frac{k}{2}} \lambda \varphi^{k-2}}, \quad (41)$$

$$m_\varphi^{(T)} = \frac{\sqrt{3}}{2} \sqrt{6^{-\frac{k}{2}} k(k-1) \alpha^{-\frac{k}{2}} \lambda \varphi^{k-2}}. \quad (42)$$

For all $k > 2$, the inflaton mass *vanishes* at the minimum as $\varphi^{(k-2)/2} \rightarrow 0$, but these expressions reduce to the more familiar nonzero masses for the quadratic case ($k = 2$): $m_\varphi^{(E)} = \sqrt{\lambda/\alpha}$ and $m_\varphi^{(T)} = \frac{1}{2}\sqrt{\lambda/\alpha}$.

One can show that in the large- N_* limit, the generalized E-models predict

$$n_s \simeq 1 - \frac{2}{N_*} - \frac{3\alpha(2+k-2\log[\frac{2kN_*}{3\alpha}])}{2kN_*^2}, \quad (43)$$

$$r \simeq \frac{12\alpha}{N_*^2} - \frac{36\alpha^2 \log[\frac{2kN_*}{3\alpha}]}{kN_*^3}, \quad (44)$$

and the generalized T-models predict

TABLE II. Predictions for the scalar spectral index n_s and tensor-to-scalar ratio r in E-models and T-models for $\alpha = 1$ and $N_* = 55$, as a function of the parameter k .

k	E-model		T-model	
	n_s	r	n_s	r
2	0.9649	0.00355	0.9638	0.00387
4	0.9643	0.00371	0.9636	0.00390
6	0.9641	0.00378	0.9636	0.00391
8	0.9639	0.00381	0.9636	0.00391
10	0.9638	0.00383	0.9636	0.00391

$$n_s \simeq 1 - \frac{2}{N_*} - \frac{3(k-2)\alpha}{2kN_*^2}, \quad (45)$$

$$r \simeq \frac{12\alpha}{N_*^2} - \frac{36\alpha^2}{kN_*^3}. \quad (46)$$

Note that the normalization of λ in Eq. (38) remains independent of k . Generalizations of the analytical expressions for φ_{end} and φ_* including the dependence on both α and k are provided in Appendix B, Eqs. (B11)–(B14).

We summarize the values of the CMB observables n_s and r predicted for $N_* = 55$ e -folds and $\alpha = 1$ in Table II. These values are exact, calculated by numerically solving for the primordial spectra, as discussed in Appendix C. We observe that, for fixed N_* , n_s decreases slightly as k increases. However, the key insight is that models with $k > 4$ can accommodate a larger number of e -folds due to modified reheating dynamics. The equation of state parameter during reheating becomes $w_\varphi = (k-2)/(k+2)$, corresponding to a stiffer equation of state. This allows for higher values of N_* , which can more than compensate for the direct k -dependence and yield larger n_s values.⁶ Models with $k \geq 6$ can thus potentially provide better agreement with ACT DR6 constraints ($n_s \simeq 0.974$) while maintaining successful predictions for r , provided the allowed range of N_* is sufficiently extended. We analyze these trade-offs in detail in Sec. VII.

These generalized attractors can arise naturally in no-scale supergravity. The supergravity construction, detailed in Appendix A, yields potentials of the form (39) and (40) through appropriate choices of the superpotential W .

B. Deformed no-scale attractors

We can further generalize the attractor models by adding in a deformation parameter κ . Such deformations lift the plateau nature of the potential at large field value and can have dramatic consequences for the CMB observables, most notably for n_s [19,32,74]. These too can be

⁶However, fragmentation effects during preheating impose an upper limit $N_{*,\text{max}}$, which we discuss in Sec. VII.

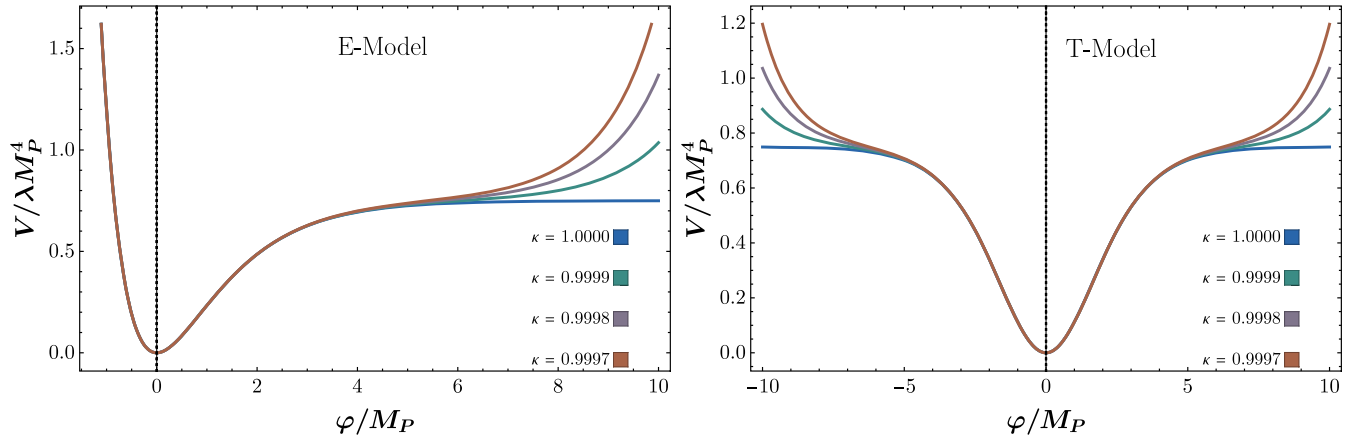


FIG. 1. Deformed attractor potentials with $\alpha = 1$ and $k = 2$. Left panel: deformed Starobinsky (E-model) potential for $\kappa = 1, 0.9999, 0.9998$, and 0.9997 . Right panel: deformed T-model potential for the same κ values. The deviations from $\kappa = 1$ introduce observable modifications to the inflationary plateau, and become substantial for $\varphi/M_P \gtrsim 6$.

constructed within the no-scale supergravity framework. The model building aspects are detailed in Appendix A.

We first consider the modified Starobinsky model with potential

$$V = \frac{3}{4} \lambda M_P^4 \left(\kappa - \kappa \cosh \left(\sqrt{\frac{2}{3\alpha}} \varphi \right) + \sinh \left(\sqrt{\frac{2}{3\alpha}} \varphi \right) \right)^k, \quad (47)$$

which reduces to the generalized α -Starobinsky model (39) when $\kappa = 1$.⁷ For κ slightly below unity, this modification introduces a small deviation from the standard attractor behavior while preserving the overall inflationary dynamics. In Sec. VII, we examine a representative value: $\kappa = 0.9999$. This small deviation from unity generates observable effects that bring the predictions into excellent agreement with the ACT DR6 data, particularly by shifting the spectral index n_s and tensor-to-scalar ratio r along trajectories that intersect the observational contours [32].

Similarly, one can construct modified T-model attractors with potential

$$V = \frac{3}{16} \lambda M_P^4 \left(1 + \kappa - (\kappa - 1) \cosh \left(\sqrt{\frac{2}{3\alpha}} \varphi \right) \right)^2 \times \tanh^k \left(\frac{\varphi}{\sqrt{6\alpha}} \right), \quad (48)$$

which reduces to the generalized T-model in Eq. (7) for $\kappa = 1$. As with the modified Starobinsky model, we investigate the same value of $\kappa = 0.9999$. These modifications preserve the attractor nature of the models while

introducing controlled deviations that enhance compatibility with current observations.

Figure 1 illustrates the evolution of the potential shapes for both model classes as κ varies. The modifications introduce subtle but phenomenologically significant changes to the inflationary plateau region, which directly impact the predicted observables.

VI. REHEATING CONSTRAINTS AND DARK MATTER PRODUCTION

Following inflation, for $k = 2$ the Universe enters a matter-dominated phase as the inflaton begins to oscillate harmonically about its minimum. For larger k , the equation of state is $w = (k - 2)/(k + 2)$ and the oscillations are in general anharmonic. The reheating process begins almost immediately after inflation ends. If the inflaton is allowed to decay into Standard Model particles with rate Γ_φ , the decay products thermalize [99–106] and the temperature of this dilute plasma quickly reaches a maximum value [107,108]

$$T_{\max} = \left[\frac{30}{g_{\max} \pi^2} \frac{\sqrt{3}}{4} \left(\frac{3}{8} \right)^{\frac{3}{2}} \Gamma_\varphi \rho_{\text{end}}^{\frac{1}{2}} M_P \right]^{1/4}, \quad (49)$$

for $k = 2$.⁸ As the inflaton continues to decay, the temperature drops from its maximum as $T \propto a^{-3(k-1)/(2k+4)}$ for decays to fermions and $T \propto a^{-3/(2k+4)}$ for decays to scalars ($T \propto a^{-3/8}$ for $k = 2$ for decays to both fermions and bosons) until reheating,⁹ when the energy density in radiation starts to dominate the expansion, and subsequently

⁸For more general expressions that depend on k and the spin of the final-state decay products, see Ref. [97].

⁹If the reheating temperature is sufficiently low, the self-fragmentation of the inflaton will occur before the end of reheating, and the evolution of the instantaneous temperature depends on the decay of free inflaton quanta, see Sec. VII.

⁷The case $\alpha = 1$, $k = 2$ was considered in [19] and more recently in [32], where κ was denoted by λ .

$T \propto a^{-1}$ for all k [97,109]. The reheating temperature for $k = 2$ is given in Eq. (37).

In the Starobinsky model derived from Eq. (1), reheating is most naturally achieved by including the Standard Model (SM) in the action. Then, after the conformal transformation to the Einstein frame, there is a non-negligible coupling of the inflaton to the Higgs kinetic term leading to a decay rate, $\Gamma_\phi \propto M^3/M_P^2$, and a reheating temperature of $\simeq 3 \times 10^9$ GeV in the SM and $\simeq 10^{10}$ GeV in the MSSM, assuming that the Higgs field is minimally coupled to gravity. This decay channel vanishes for a coupling $\xi|H|^2 R$ with $\xi = \frac{1}{6}$. However, in that case the inflaton can still decay to SM gauge bosons through the trace anomaly [110], which results in a reheating temperature of order 10^8 GeV.

In the supergravity formulation of the Starobinsky and related attractor models, additional SM chiral superfields can be added inside the logarithm of Eq. (8). In that case, when one transforms to a canonical inflaton, its couplings with the SM fields appear as if they were coupled conformally to curvature with $\xi = \frac{1}{6}$ [77,79]. Thus, barring a direct superpotential coupling between the inflaton and Standard Model fields, the decays of the inflaton are highly suppressed [111,112]. Nevertheless, we expect that in this case as well inflaton decay should proceed through the coupling of the Kähler potential to the trace anomaly [113]: $\mathcal{L} \supset (-K/6)T_\mu^\mu$. However for ϕ -inflation [see Eq. (10)], $\langle \partial K / \partial \phi \rangle = 0$ and this contribution vanishes as well. Alternatively, if the gauge kinetic function depends on the inflaton, decays to gauge bosons and gauginos may be the dominant source for reheating [111,112]. In what follows, we will not tie ourselves to any particular mode of decay, and simply treat the reheating temperature, T_{RH} as a (relatively) free quantity, subject only to general phenomenological constraints.

We have already commented that we consider $T_{\text{RH}} = 4$ MeV as a lower bound on the reheating temperature to ensure standard BBN [73]. We nevertheless consider $T_{\text{RH}} \gtrsim 100$ GeV as a reasonable lower bound to ensure the generation of a baryon asymmetry. Below the electroweak scale, sphaleron interactions are out-of-equilibrium and there are no interactions which are capable of shuffling any existing lepton or baryon asymmetries, thus requiring direct baryon number violation at scales below the electroweak scale. While we know of no proof that generating a baryon asymmetry at lower energies is not possible, we consider the requirement of reheating above the electroweak scale as reasonably conservative.

For inflaton couplings approaching the nonperturbative limit, the decay rate, $\Gamma_\phi \sim H_I$, where the Hubble parameter H_I during inflation is $H_I = M/2 \sim \sqrt{\lambda} M_P/2$ for the Starobinsky model. From Eq. (37), this leads to $T_{\text{RH}} \sim 2 \times 10^{15}$ GeV. In supersymmetric models, high reheating temperatures generally lead to the overproduction of gravitinos which in turn lead to the overproduction of

dark matter if the gravitino is not the lightest supersymmetric particle [71,108,114].

We can approximate the thermal production of gravitinos as [84,108,115–121]

$$Y_{3/2} \equiv \frac{n_{3/2}}{n_{\text{rad}}} \simeq 0.003 \left(\frac{\Gamma_\phi}{M_P} \right)^{1/2}, \quad (50)$$

where $Y_{3/2}$ is the gravitino yield, $n_{\text{rad}} = \zeta(3)T^3/\pi^2$. If gravitinos decay after the lightest supersymmetric particle (LSP) freezes out, their decay products contribute to the LSP abundance and we have the constraint

$$\Omega_{\text{LSP}} h^2 = m_{\text{LSP}} Y_{3/2} \left(\frac{n_\gamma}{2\rho_c} \right) \leq 0.12. \quad (51)$$

Using Eq. (37), we can translate this into a limit on the reheating temperature, $T_{\text{RH}} \lesssim 10^{10}$ GeV. Consequently, in SUSY realizations subject to this bound, the standard $\alpha = 1$ no-scale attractors are disfavored by ACT DR6 (while remaining broadly consistent with Planck), motivating the generalized/deformed models studied below.

These considerations, combined with the CMB constraints from ACT DR6, significantly restrict the viable parameter space. In supersymmetric realizations, radiative corrections to the inflaton potential are suppressed by supersymmetry [122–124], protecting the flatness required for successful inflation.

VII. RESULTS

We now present the constraints on attractor models of inflation we obtain using the *Planck* 2018 [3], BICEP/Keck 2018 [4], ACT DR6 [10], and SPT-3G [12] CMB data. Our analysis in the next subsection reveals significant tension between the canonical Starobinsky model (with $\alpha = 1$) and ACT DR6. There remains general agreement with the *Planck* 2018 data, but there is already some tension with the SPT-3G data. These tensions are reduced for $\alpha > 1$, and we show in subsequent subsections that the tensions are further relaxed in generalized attractor models with $k > 2$ as well as in deformed models with $\kappa < 1$. In all cases we base our analysis on the exact numerical values of the CMB observables, computed as discussed in Appendix C.

A. Standard attractor models

As discussed in Sec. IV, the number of e -folds and φ_* depend on the reheating temperature, which appears in Eq. (27) through ρ_{rad} . This in turn induces a dependence of n_s on T_{RH} . In Fig. 2, we show this dependence for the E- and T-models (upper and lower panels) for $\alpha = 1$ and 10 (left and right panels). We note that although both N_* (defined at the pivot scale of 0.05 Mpc^{-1}) and n_s depend on T_{RH} , the relation between N_* and n_s has some model dependence. The horizontal shading in Fig. 2 shows the

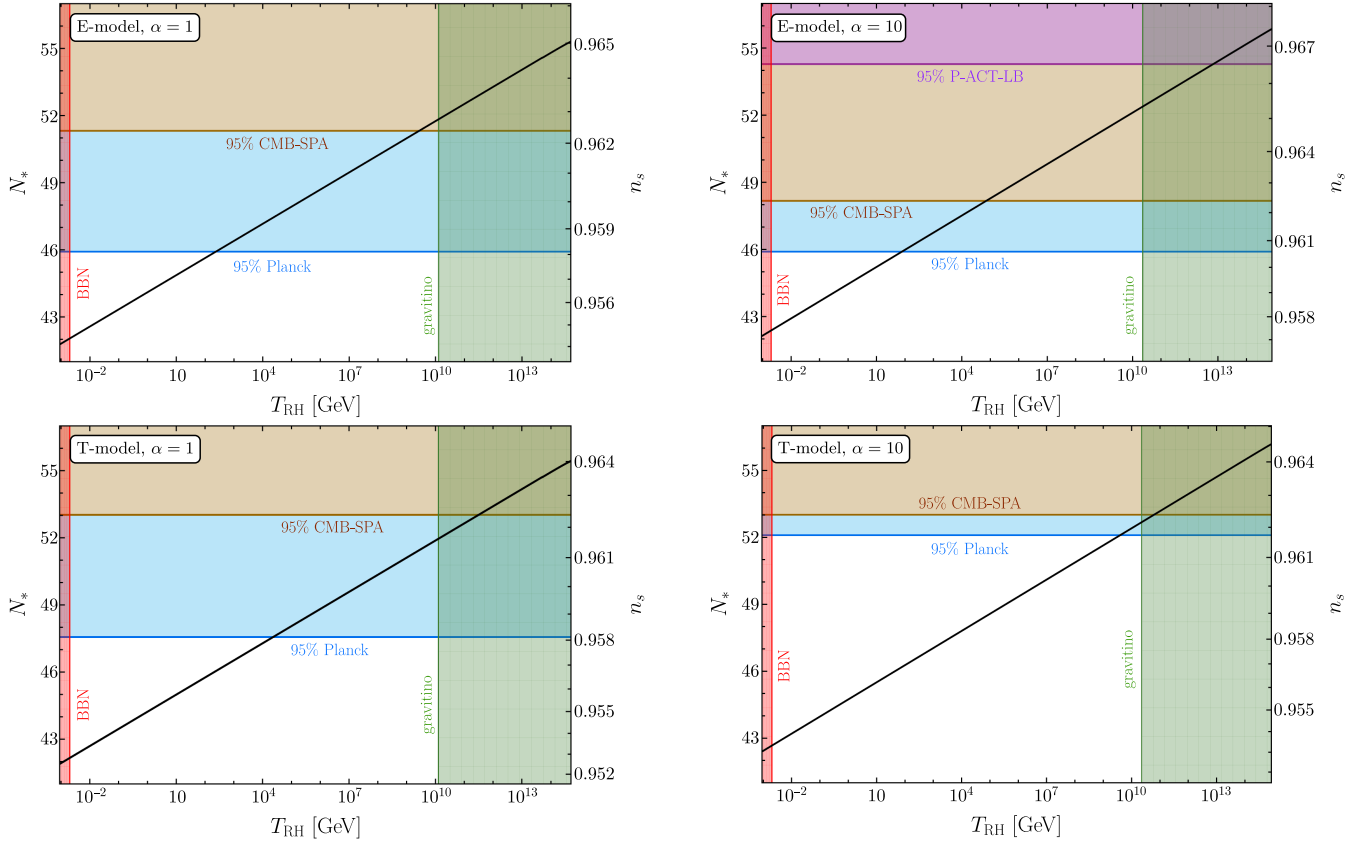


FIG. 2. The relation between N_* (defined for a pivot scale of 0.05 Mpc^{-1}) and the reheating temperature in the E-model (upper panels) and the T-model (lower panels), for $\alpha = 1$ (left panels) and $\alpha = 10$ (right panels). The horizontal shadings corresponds to the 95% CL bounds from *Planck* 2018, CMB-SPA, and P-ACT-LB. The vertical shadings show the constraints of T_{RH} from big bang nucleosynthesis (BBN) and gravitino production in supersymmetric models. Note that the limits on n_s (right vertical axis) depend on α , as the calculated value of r depends on α as seen in Eq. (46).

95% CL lower limit on n_s (blue for the limit from *Planck* 2018 alone, brown for the SPT-*Planck*-ACT combination (CMB-SPA), and purple for *Planck*-ACT-DESI combination (P-ACT-LB) seen only in the upper right panel for the E-model with $\alpha = 10$). The vertical shading shows exclusions of reheating temperatures below 4 MeV from BBN on the left side of each panel, and on the right side temperatures above $\simeq 10^{10}$ GeV due to the gravitino bound discussed above. For $\alpha = 1$ there is a range of reheating temperatures consistent with the *Planck* 95% lower bound on n_s ($220 - 10^{10}$ GeV) for the E-model and ($2 \times 10^4 - 10^{10}$ GeV for the T-model). Neither model falls within the P-ACT-LB 95% range for any possible reheating temperature. The limits on n_s depend on r as can be seen in Fig. 3, and r depends on α as seen in Eq. (46). Thus limits on n_s are model-dependent and are slightly different in each panel of Fig. 2. However, for $\alpha = 10$ in the E-model, $T_{RH} \gtrsim 80$ or 10^5 GeV are allowed by *Planck* 2018 and CMB-SPA respectively. The P-ACT-LB bound is now visible but requires $T_{RH} \gtrsim 10^{13}$ GeV, in excess of the gravitino bound, but still less the temperature attained in instantaneous reheating. For the T-model with $\alpha = 10$, *Planck* is satisfied

for $T_{RH} \gtrsim 5 \times 10^9$ GeV, but CMB-SPA cannot be satisfied simultaneously with the gravitino bound.

Figure 3 displays the observational constraints on the α -attractor models in the (n_s, r) plane. We overlay the 68% and 95% CL contours from *Planck*/BICEP/Keck [4] (blue shadings), the P-ACT-LB combination [11] (purple contours) and the CMB-SPA dataset [12] (brown rectangles). Note that the *Planck* contours are provided at the WMAP pivot scale $k_* = 0.002 \text{ Mpc}^{-1}$, while the P-ACT-LB (and our predictions) correspond to $k_* = 0.05 \text{ Mpc}^{-1}$ for both n_s and r . CMB-SPA does not provide an r -dependent limit on n_s . The theoretical predictions show the trajectories for both the α -Starobinsky (E-model) and T-model potentials given by Eqs. (6) and (7), respectively. The left panel shows the E-model predictions for three highlighted values of $\alpha = 1, 10$, and 25 . The solid line trajectories correspond to specific reheating temperatures of 4 MeV (labeled BBN), 100 GeV (labeled T_{EW} , 10^{10} GeV (as labeled) corresponding to the gravitino bound, and 2×10^{15} GeV corresponding to instantaneous reheating (labeled as $\Gamma_\phi = H$). The red shading highlights the range $10^2 \text{ GeV} < T_{RH} < 10^{10}$ GeV. For reference, we also show the trajectory for $N_* = 50$

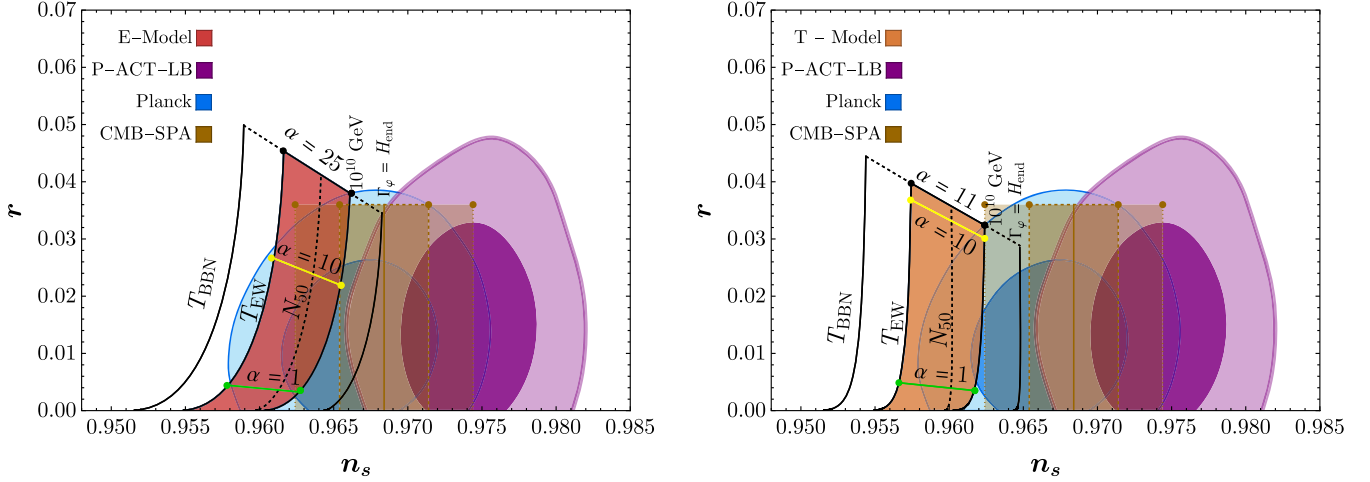


FIG. 3. Constraints on α -attractor models showing the 68% and 95% CL contours from *Planck*/BICEP/Keck [4] (blue shadings), the P-ACT-LB combination [11] (purple contours) and the CMB-SPA dataset [12] (brown rectangles). Note that the latter provides only an r -independent limit on n_s . All pivot scales are taken as $k_* = 0.05 \text{ Mpc}^{-1}$ except for the *Planck* pivot scale for r , conventionally chosen to be $k_* = 0.002 \text{ Mpc}^{-1}$. Left panel: α -Starobinsky (E-model) predictions in the (n_s, r) plane. Solid lines indicate reheating temperatures from T_{BBN} (4 MeV), T_{EW} (100 GeV), 10^{10} GeV (the gravitino bound), and $2 \times 10^{15} \text{ GeV}$ (instantaneous reheating with $\Gamma_\varphi = H$). The dashed line shows $N_* = 50$ for reference. Right panel: as in the left panel for the T-model predictions.

e -folds before the end of inflation. The right panel shows similar results for the T-model with $\alpha = 1, 10$, and 11 as examples.

Results for the canonical Starobinsky model (E-model with $\alpha = 1$) are shown in the left panel of Fig. 3. It predicts $n_s \simeq 0.961$ for $N_* = 50$, corresponding to a reheating temperature $T_{\text{RH}} \simeq 5 \times 10^7 \text{ GeV}$, which lies comfortably within the *Planck*+BICEP/Keck 95% confidence region, but below the 95% CL range of the CMB-SPA dataset, and falls significantly below the ACT DR6 preferred value of $n_s = 0.9752 \pm 0.0030$. The model remains observationally viable for $\alpha \lesssim 25$, beyond which it violates the upper limit on r for $T_{\text{RH}} \leq 10^{10} \text{ GeV}$. Efficient reheating (though in violation of the gravitino bound) maximizes N_* and pushes n_s toward higher values, improving agreement with ACT DR6 (purple shading). The right panel presents the corresponding constraints for T-models. These models predict systematically lower n_s values than E-models for the same α , with the canonical case ($\alpha = 1$) yielding $n_s \simeq 0.960$ for $N_* = 50$ e -folds. This places T-models in even greater tension with ACT DR6 data. The observational constraints restrict T-models to $\alpha \lesssim 11$, a more stringent limit than in the α -Starobinsky case. Both model classes require efficient reheating (high reheating temperatures) to approach the region preferred by the ACT DR6 data.

B. Generalized attractor models

We next examine generalized α -attractor models with nonquadratic minima of the form $V \propto \varphi^k$, which modify both the inflationary dynamics and reheating phase. These models provide enhanced flexibility in reconciling theoretical predictions with the latest CMB observations,

particularly the higher spectral index values favored by ACT DR6. The generalized E-model attractors are described by Eq. (39) and the generalized T-models are described by Eq. (40). The modification from the standard quadratic minimum ($k = 2$) to higher powers significantly alters the reheating dynamics through the effective equation of state $w_\varphi = (k - 2)/(k + 2)$, leading to enhanced e -fold numbers for fixed reheating temperatures. This effect is illustrated in Fig. 4, which shows the dependence of N_* (defined at the pivot scale $k_* = 0.05 \text{ Mpc}^{-1}$) for $k = 2, 4, 6, 8$, and 10 . Solid lines correspond to the E-models and dashed lines correspond to the T-model, though the difference between the two is generally very small on the scale of this figure. The lines for $k = 2$ are identical to the ones given in Fig. 2 for $\alpha = 1$. All of the lines converge when reheating is instantaneous, as the only effect from the shape of the potential is limited to the difference in the energy densities at the end of inflation, which is weakly dependent on k (see Eqs. (B11) and (B12) in Appendix B). We note that $w_\varphi = 1/3$ for $k = 4$, and the contribution to N_* from ρ_{rad} in Eq. (27) drops out. Therefore, $N_* \simeq 56$ is mostly independent of T_{RH} , except for the indirect dependence through g_{RH} .

For $k > 4$, $w_\varphi > 1/3$, the sign of the contribution from ρ_{rad} in Eq. (27) changes, and N_* increases with decreasing T_{RH} . However, in such a case, the self-interaction of the inflaton cannot be ignored, as it will drive the resonant growth of inflaton inhomogeneities, eventually fragmenting the inflaton condensate. The net effect is a transition to a radiation-dominated epoch, $w_\varphi \rightarrow 1/3$, where the dominant species are the nearly massless inflaton quanta [125–129]. The timescale for fragmentation is fixed for given values of λ, k, α , and can be smaller than the reheating timescale, when

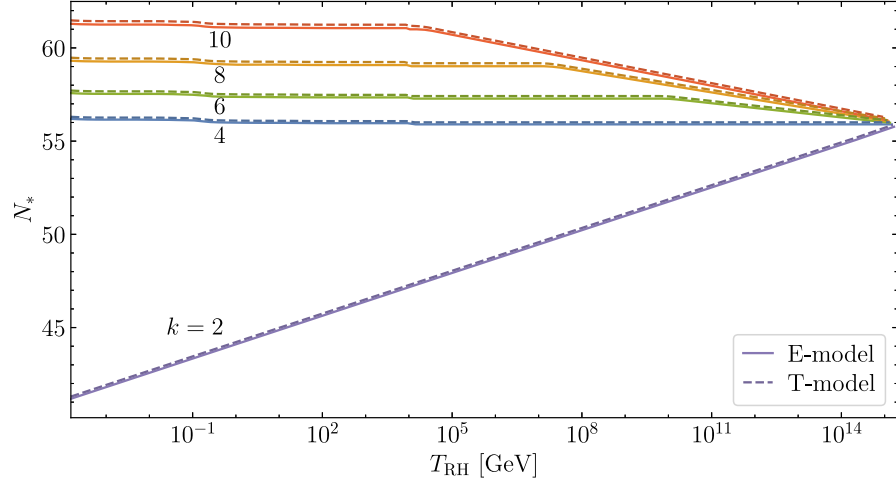


FIG. 4. Plot of N_* as a function of T_{RH} for different values of k in the E- and T-models with $\alpha = 1$.

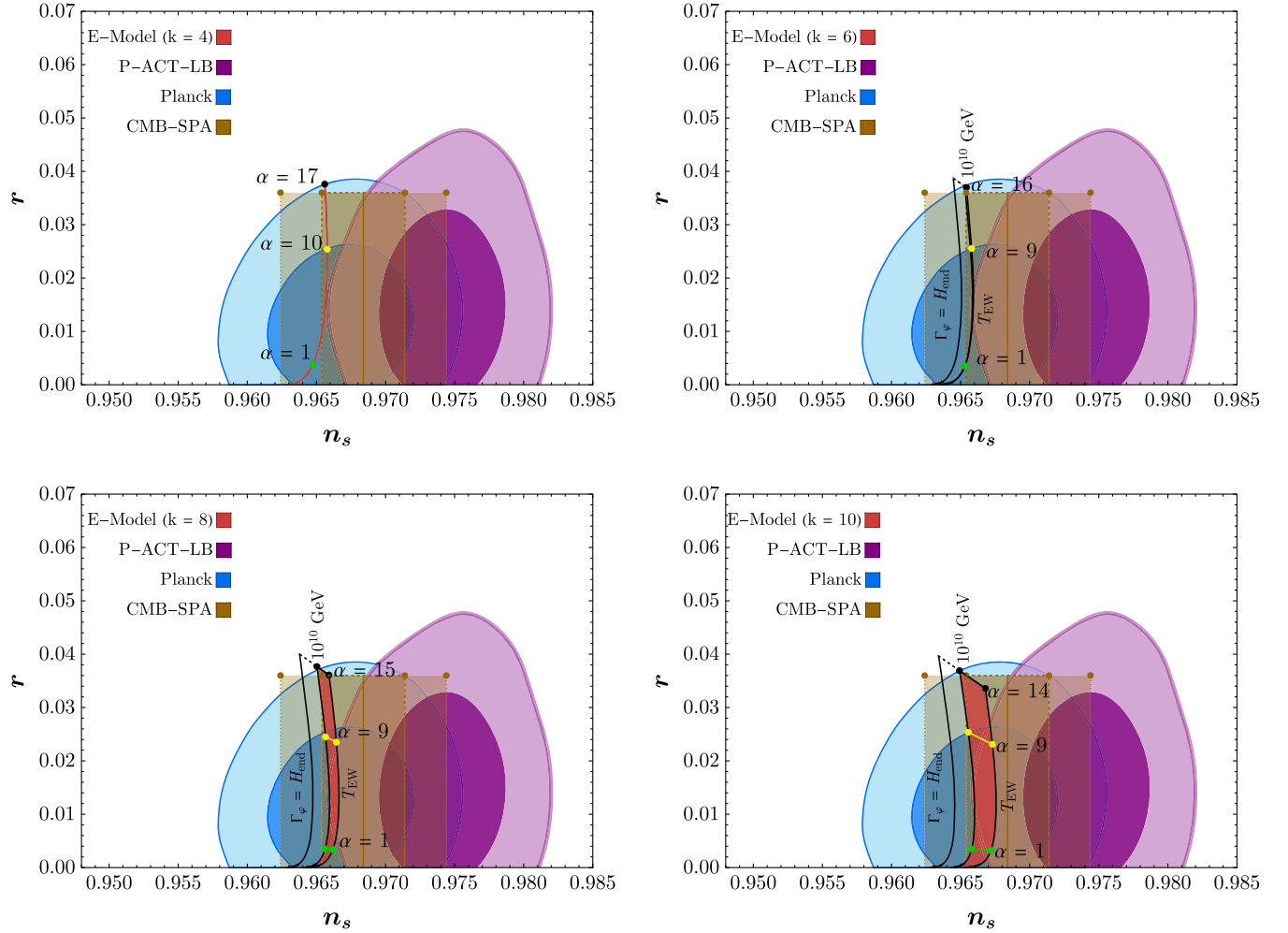


FIG. 5. As in Fig. 3, showing the constraints on generalized α -Starobinsky (E-model) attractor models with $V \propto \varphi^k$ minima for $k = 4, 6, 8$, and 10 . Shaded bands indicate the range of N_* allowed by reheating temperatures from T_{EW} to 10^{10} GeV. The curve for T_{BBN} is degenerate with that shown for T_{EW} when fragmentation effects are included (see Fig. 4). For $k > 4$, the curve for instantaneous reheating is to the left of the shaded band. Higher k values systematically shift predictions toward larger n_s , improving consistency with the ACT DR6 data. The constraint on r limits all models to $\alpha \lesssim 14\text{--}17$.

the now free inflaton quanta complete their decay into radiation. Read from right to left, Fig. 4 follows the increasing trend of N_* as a function of T_{RH} for $k > 4$, until low reheating temperatures, for which reheating is completed after fragmentation, and the dependence of N_* on T_{RH} becomes flat (see Eq. (B15) in Appendix B).

Figure 5 presents constraints on generalized E-model attractors given by Eq. (39) for $k = 4, 6, 8$, and 10. For $k = 4$ (upper left panel), the predictions track closely those of the standard E-model, with the observational constraints permitting $\alpha \lesssim 17$. Note that, in this case, there is no theoretical band, as there is no dependence of n_s (and r) on T_{RH} . Therefore we see only a single curve which varies with α . The canonical case ($\alpha = 1$) predicts $n_s \simeq 0.965$ for the allowed e -fold range. This runs directly through the *Planck* 2018 ellipses and is also quite compatible with the CMB-SPA dataset. It is barely consistent with the 95% bound from the P-ACT-LB dataset when $\alpha \simeq 5$.

As k increases to 6 (upper right panel), there is only a narrow allowed range for N_* between $\simeq 56$ –58. Note that in contrast to $k = 2$, lower values of T_{RH} correspond to the right side of the trajectory strip. As for $k = 4$, there is no shaded strip as the trajectories for $T_{\text{RH}} = 100$ GeV and 10^{10} GeV are nearly identical (as is the trajectory for $T_{\text{RH}} = 4$ MeV) and these three trajectories in the upper right panel of Fig. 5 are nearly degenerate. As can be seen in Fig. 4, the value of N_* for instantaneous reheating is somewhat lower and yields the left trajectory in that panel. Concerning n_s and r , we observe more substantial shifts: at $\alpha \simeq 5$ the prediction intersects the ACT DR6 95% confidence region at $N_* \simeq 58$, corresponding to $n_s \simeq 0.966$. The canonical model ($\alpha = 1$) yields $n_s \simeq 0.965$, representing a notable enhancement compared to the standard ($k = 2$) case. The upper limit on α is slightly lowered to $\alpha < 16$. The trend continues for $k = 8$ and 10 (lower panels), where the predictions systematically shift toward higher n_s values. For these values of k , there is a slight difference in N_* for $T_{\text{RH}} = T_{\text{EW}}$ and 10^{10} GeV, resulting in the thin shaded strip seen in the lower two panels. The trajectory for $T_{\text{RH}} = 4$ MeV remains degenerate with the one for $T_{\text{RH}} = T_{\text{EW}}$. For $k = 10$ with $\alpha \simeq 5$, the model predictions overlap substantially with both the ACT DR6 and CMB-SPA preferred regions. The maximum viable α remains approximately constant at $\alpha \sim 16$ for all k values, determined by the tensor-to-scalar ratio constraint $r \lesssim 0.04$.

Figure 6 displays the corresponding analysis for the generalized T-model attractors given by Eq. (40). While T-models inherently predict lower spectral indices than their E-model counterparts, the higher- k variants demonstrate significantly improved agreement with observations. For $k = 4$ (upper left), $\alpha \lesssim 11$ yields predictions that are fully consistent with *Planck* 2018, within the 95% CL of CMB-SPA, but remain below the ACT DR6 values. However, increasing to $k = 6$ (upper right) allows N_* to increase and shifts the trajectory closer to the P-ACT-LB observational

contours, with n_s reaching ~ 0.965 for $\alpha = 1$ at $N_* \simeq 58$ e -folds. As for the E-model, the trajectories for $T_{\text{RH}} = 10^{10}$ GeV, 100 GeV, and 4 MeV are nearly degenerate. The most dramatic improvement occurs for $k = 8$ and 10 (lower panels). For $k = 10$ with $\alpha \simeq 1$, the model predictions enter the ACT DR6 95% confidence region, reaching $n_s \simeq 0.965$ –0.967 over the allowed e -fold range of $N_* \simeq 59$ –61.5 for $T_{\text{RH}} = 100$ – 10^{10} GeV.

C. Deformed no-scale attractors

We examine next deformed α -attractor models, introducing a parameter $\kappa < 1$ that controls this deviation as in Eq. (47) for the E-model and in Eq. (48) for the T-model. These modifications preserve the attractor nature of the models while providing enhanced flexibility to mitigate the tensions with different CMB datasets. Figure 7 presents the modified E-model (left) and T-model (right) predictions for $\kappa = 0.9999$. This minimal deviation from the standard attractor ($\kappa = 1$) creates a distinctive trajectory that passes directly through the centers of the observational confidence regions. For the E-model and the canonical case ($\alpha = 1$) with $\kappa = 0.9999$, the model predicts $n_s \simeq 0.968$ –0.975 and $r \simeq 0.007$ –0.005 across the reheating range from T_{EW} to 10^{10} GeV. This represents a substantial shift from the standard Starobinsky prediction. The value of n_s is now *high* compared with both *Planck* 2018 and CMB-SPA but is now in agreement with P-ACT-LB preferred range. The modification preserves the fundamental attractor structure, evidenced by the convergence of different α trajectories at large N_* , while introducing sufficient flexibility to reconcile diverse observational constraints. As discussed in the previous section and Appendix A, such modifications arise naturally within the no-scale supergravity framework through appropriate choices of the Kähler potential and superpotential.

Similar results are found for the deformed T-model, as shown in the right panel of Fig. 7. For the same value of $\kappa = 0.9999$ and $\alpha = 1$, the shift in n_s is less extreme giving $n_s \simeq 0.970$ –0.977 for the same range in reheating temperatures of T_{EW} to 10^{10} GeV, which is, within uncertainties, in agreement with all three datasets considered. Similar effects on n_s in deformed models were seen in [74]. Note that for $\alpha > 1$, the effect of the deformation is minimal and the range in n_s is similar to that found in the undeformed attractor models. We now find upper limits of $\alpha \leq 9$ (E-model) and $\alpha \leq 11$ (T-model).

Figure 8 extends this analysis to include generalized deformed attractors with nonquadratic minima. The top panels display modified E-models with $\kappa = 0.9999$ for $k = 4$ (left) and $k = 6$ (right). For $k = 4$ with $\alpha = 1$, the model predicts $n_s \simeq 0.970$ and $r \simeq 0.004$. These models achieve exceptional agreement with the *Planck* 2018, CMB-SPA, and P-ACT-LB datasets, demonstrating that perturbations of order 10^{-4} to the attractor structure can fully reconcile present observational tensions. For $k = 6$, the $\alpha = 1$

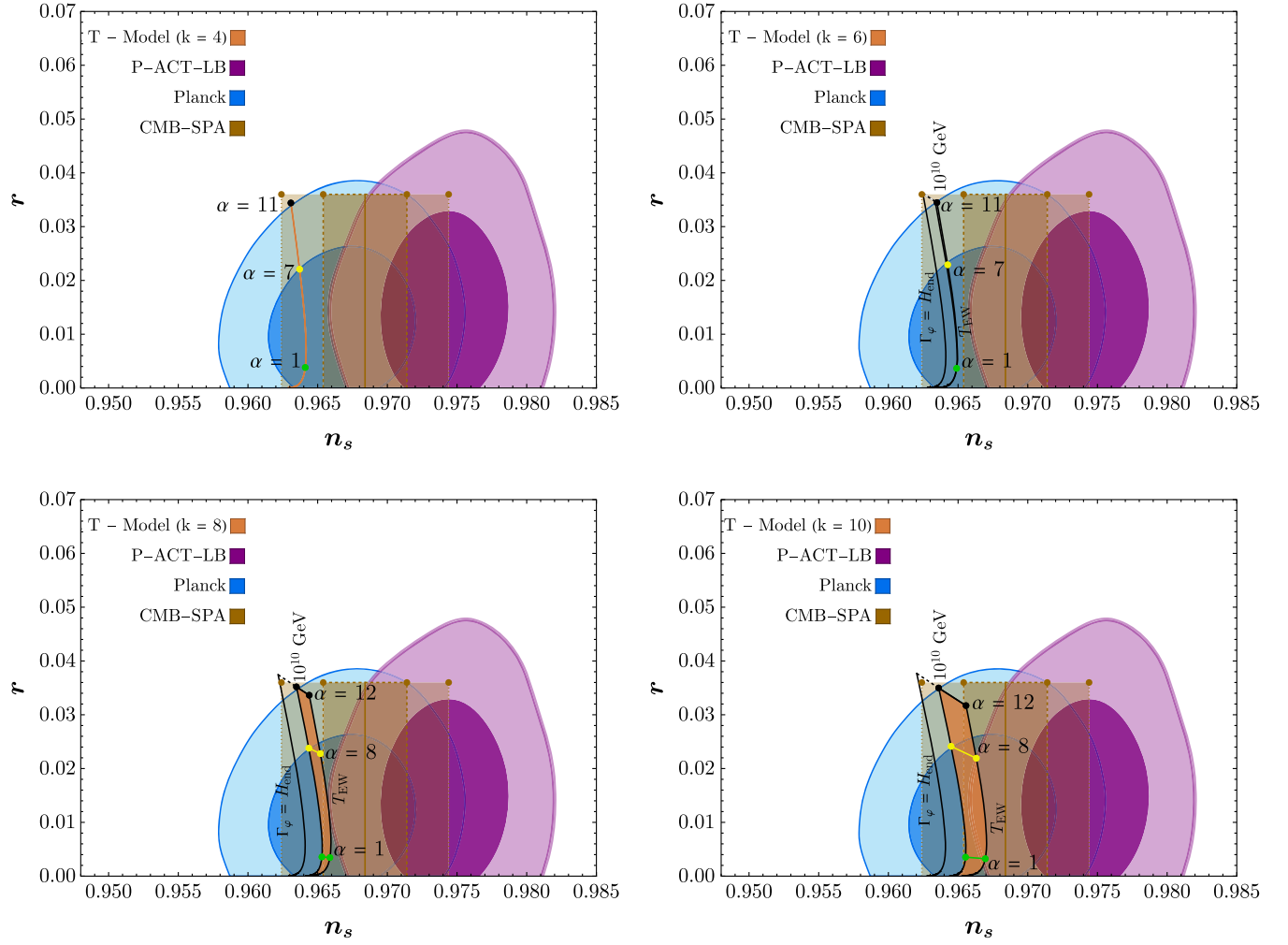


FIG. 6. As in Fig. 5, showing the constraints on generalized T-model attractors with $V \propto \varphi^k$ minima for $k = 4, 6, 8$, and 10 . While T-models predict systematically lower n_s than E-models, increasing k substantially improves compatibility with observations. For $k = 10$, even the canonical model with $\alpha = 1$ approaches the observational confidence regions.

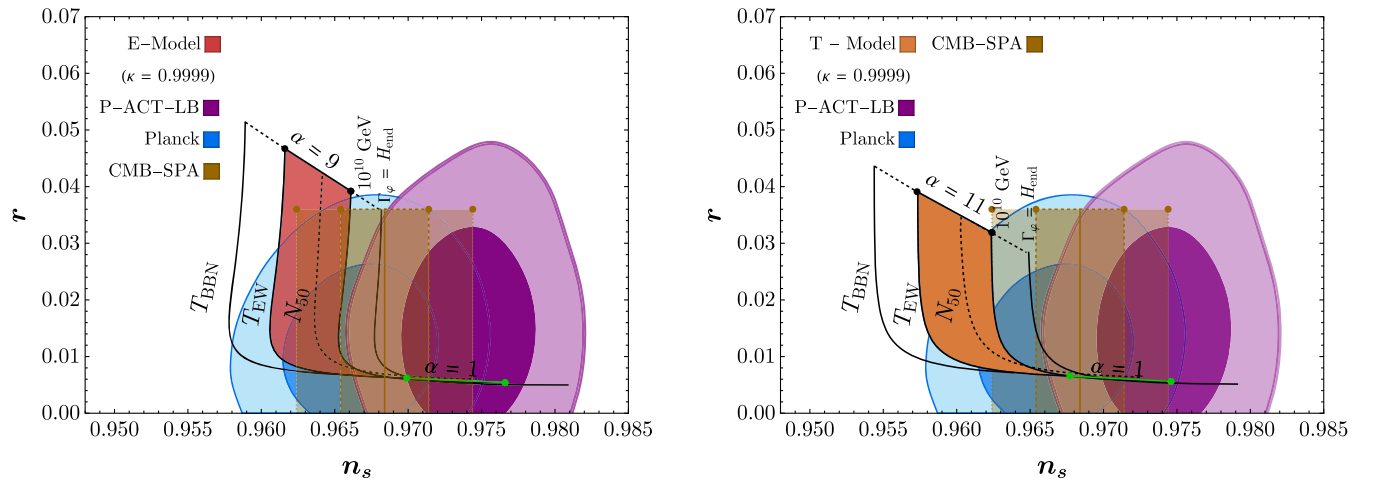


FIG. 7. As in Fig. 3, showing the constraints on deformed E-model attractors (47) (left panel) and T-model attractors (48) (right panel) with the deformation parameter $\kappa = 0.9999$. Observational contours and theoretical trajectories for various α values demonstrate how minimal modifications ($|1 - \kappa| \sim 10^{-4}$) shift predictions through the centers of observational confidence regions. The canonical model ($\alpha = 1$) with $\kappa = 0.9999$ achieves remarkable agreement with the P-ACT-LB dataset while maintaining consistency with the r constraint. Red-shaded band (E-model) and orange-shaded band (T-model) indicate reheating uncertainty from T_{EW} to 10^{10} GeV.

prediction is high relative to *Planck* 2018 and CMB-SPA, but agreement with all three datasets is achieved for slightly higher $\alpha \leq 4$. Indeed, $k = 6$ yields the prediction $n_s \simeq 0.974$ with $r \simeq 0.016$ for $\alpha = 4$ and $N_* = 56\text{--}58$ e -folds. The upper limits on α from *Planck* 2018 are $\alpha \leq 15$ for $k = 4$ and $\alpha \leq 16$ for $k = 6$.

The bottom panels of Fig. 8 present the corresponding modified T-model predictions. For the same value of κ , the shift in n_s is smaller for the T-models. For $k = 4$ with $\kappa = 0.9999$ and $\alpha = 1$, we find $n_s \simeq 0.966$ and $r \simeq 0.004$, which satisfies the CMB-SPA constraint but remains marginally outside the P-ACT-PB 95% confidence region. However, for $k = 6$ case the shift in n_s is more substantial, leading to better concordance with the observations. In this case, for $\alpha = 1$, the predictions reach $n_s \simeq 0.969$ with $r \simeq 0.012$, entering the overlap region between all observational constraints.

To summarize our findings, we identify three exemplary scenarios for reconciling α -attractor models with current observations. (a) Standard attractors ($\kappa = 1$, $k = 2$) remain viable with $\alpha \lesssim 40$ for E-models and $\alpha \lesssim 11$ for T-models when requiring efficient reheating. These models satisfy the combined CMB-SPA constraints but show tension with ACT DR6 observations. (b) Generalized attractors ($\kappa = 1$, $k \geq 8$) naturally accommodate higher n_s values through modified reheating dynamics, achieving compatibility with all datasets for moderate α values. (c) Deformed attractors ($\kappa \simeq 0.9999$) provide the most complete reconciliation, with even canonical models ($\alpha = 1$) capable of matching the full range of observational preferences. The deformed and generalized variants provide theoretical frameworks that can accommodate the full spectrum of current observations while maintaining the appealing features of attractor models of inflation.

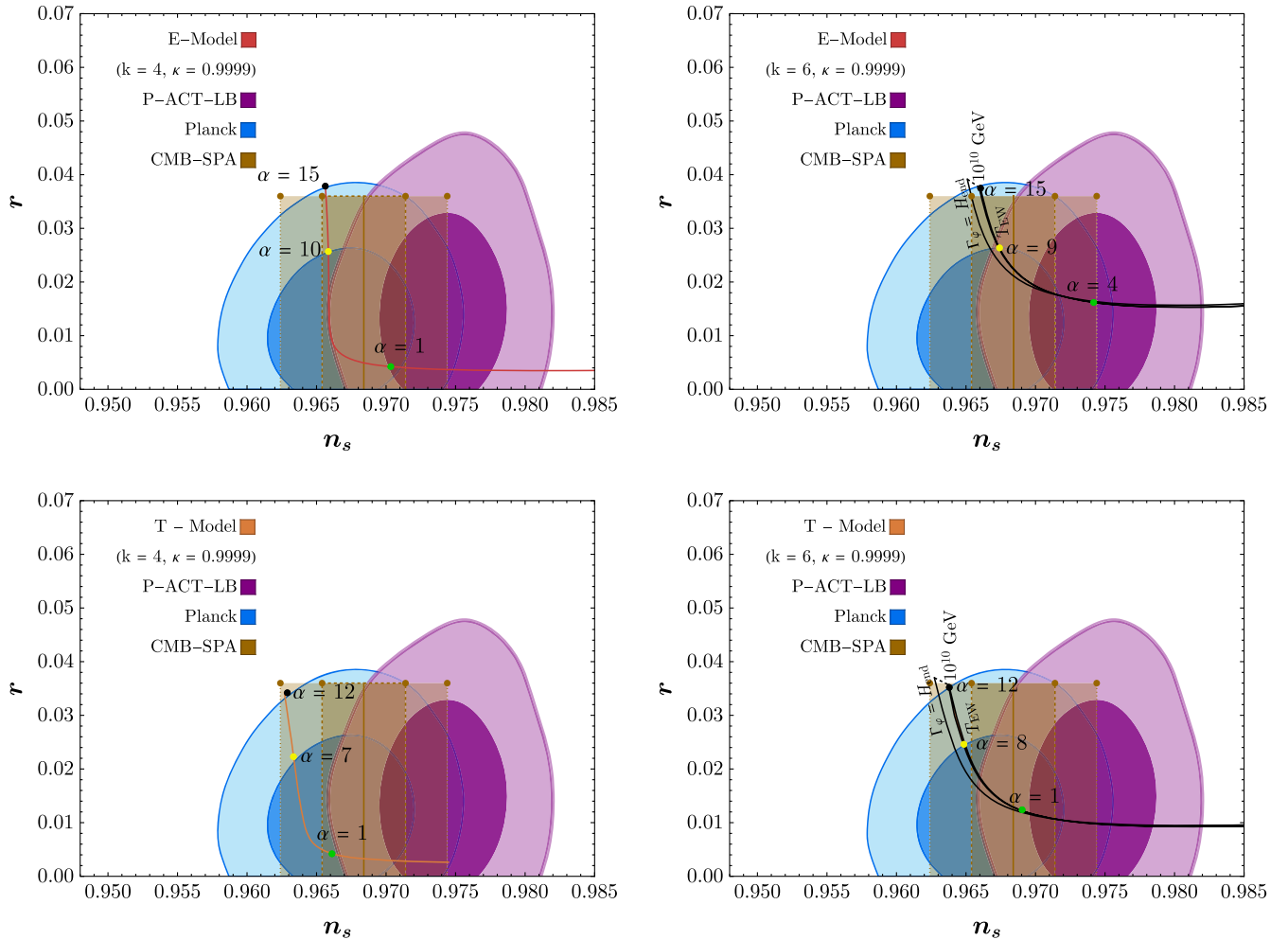


FIG. 8. As in Fig. 3, showing the constraints on deformed attractor models with $\kappa = 0.9999$ for generalized potentials with $k = 4$ (left panels) and $k = 6$ (right panels). Both E-models (upper panels) and T-models (lower panels) show significant shift in the predictions for n_s for $\alpha \sim 1$. These results demonstrate that deformations of order $|\kappa - 1| \sim 10^{-4}$ can fully reconcile attractor models with current observations.

VIII. CONCLUSIONS

Currently, models of inflation are constrained by CMB observations that determine the spectral tilt of scalar perturbations, n_s , and bound the tensor-to-scalar ratio, r . The former indicates a definite departure from purely scale-free perturbations, as is expected in many models of single-field inflation. The convergence of a value of n_s near 0.97 began with WMAP [2], and *Planck* 2018 data [3] reached a precision exceeding 0.5%. Among the well studied and relatively simple single-field models of inflation, the Starobinsky model [6] appeared to be in best agreement with *Planck* 2018 data. For example, for $N_* \sim 52$ (corresponding to a reheating temperature $T_{\text{RH}} \sim 10^{10}$ GeV), this model predicts $n_s \sim 0.963$, to be compared with the 68% CL range of 0.961–0.969 determined by *Planck*. Similar agreement is attained in the related T-models [26], which predict a slightly lower value for n_s for the same reheating temperature.

These models are special cases of a more general class of model known as α -attractors. The Starobinsky model (and its T-model cousin) can be formulated in the context of no-scale supergravity [19,81,109] with a Kähler potential given by Eq. (8) with $\alpha = 1$. The Kähler potential determines the metric in field space, and the no-scale potential characterizes a field space of constant curvature ($R = 2/3$). As is evident from the form of Eq. (8), choosing $\alpha \neq 1$ yields field spaces of arbitrary constant curvature $R = 2/3\alpha$, generating the class of α -attractor models [20–23]. Typically, models with $\alpha > 1$ predict similar (though slightly higher) values of n_s but significantly higher values of r . For example, the Starobinsky model with $N_* = 52$ predicts $r \simeq 0.0039$, to be compared with the *Planck*/BICEP/Keck upper limit of $r < 0.036$ [4], whereas this limit is saturated for $\alpha = 25$ in the E-model, and $\alpha = 11$ in the T-model.

In addition to the 1-parameter class of α -attractors generated by the prefactor of the Kähler potential in Eq. (8), we have considered generalized versions of the E- and T-models as described by Eqs. (39) and (40), respectively. These models can also be formulated within no-scale supergravity by altering the superpotential as described in Appendix A. These models have the property that the potential takes for the form $V(\varphi) \sim \varphi^k$, when expanded about the origin. The $k = 4$ models have the interesting feature that N_* does not depend on the reheating temperature and therefore the predictions of n_s and r depend only on α as shown in the upper left panels of Figs. 5 and 6.

The generalized models with $k > 2$, typically provide larger values of n_s . For example, the E-model with $k = 6$ predicts a narrow range in $n_s \sim 0.965$ in excellent agreement with *Planck* 2018 and the CMB-SPA data combination, though it still falls slightly below the 95% CL region of the P-ACT-LB combination. The apparent preference for $k > 2$ of certain datasets might indicate modified reheating dynamics, since higher k values lead to stiffer equations of state during inflaton oscillations, naturally extending the number of e -folds and raising n_s .

As discussed in [32,130], many models of inflation have an accidental nature to them, as the form of the potential may rely on a careful adjustment of some coupling to realize successful inflation. For example, the ratio of the quadratic and cubic couplings in Eq. (9) must be extremely close to $2/3\sqrt{3}$ to obtain the Starobinsky potential. Slight deformations may lead to very different predictions to n_s and r [19]. Hence we have also studied a further generalization of the E- and T-models by allowing for small deformations of the potential as given by Eqs. (47) and (48), characterized by a parameter $\kappa \simeq 1$. These too can be derived from no-scale supergravity as discussed in more detail in Appendix A. The required fine-tuning $|\kappa - 1| \sim 10^{-4}$ suggests that if such modifications exist in nature, they must arise from highly suppressed corrections, possibly from Planck-scale physics, string-theory effects, or radiative corrections.

We have found that the deformed E- and T-models can lead to significant changes in n_s , particularly for $\alpha \simeq 1$. For example, for $\kappa = 0.9999$, $k = 2$, and $\alpha = 1$, the E-model predicts $n_s \simeq 0.968$ –0.975 with $r \simeq 0.007$ –0.005 for a reheating temperature between 100 and $\sim 10^{10}$ GeV. For $\alpha > 1$, the predictions approach those of the undeformed models. For $k > 2$, similar results are found, though the range of n_s is smaller due to the lack of sensitivity to the reheating temperature. Similar effects were found in the presence of deformations caused by combining inflation with a grand unified theory such as SU(5) or SO(10) [74].

Next-generation experiments will provide definitive tests of the α -attractor paradigm. LiteBIRD will provide full-sky B-mode polarization maps with sensitivity $\sigma(r) \simeq 0.001$, enabling direct constraints on α and discrimination between E-models and T-models via their different (r, n_s) predictions [70]. If future measurements converge on $n_s \simeq 0.965$ with $r < 0.001$, this would strongly favor small- α E-models, pointing toward specific supergravity embeddings. Conversely, detection of $r \sim 0.01$ would indicate $\alpha \sim 1$ –10.

Our analysis has shown how attractor models of inflation can navigate successfully the challenges posed by increasingly precise observations of the CMB. While individual experiments show preferences for different parameter regions, the theoretical flexibility of the framework, through generalized potentials, modified attractors, and varying reheating scenarios, ensures its continued viability. As we approach the era of next-generation CMB experiments, attractor models remain among the most compelling and testable paradigms for cosmological inflation, with the potential to reveal fundamental aspects of quantum gravity and early Universe physics.

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DATA AVAILABILITY

The data that support the findings of this article are not publicly available. The data are available from the authors upon reasonable request.

APPENDIX A: GENERALIZED INFLATIONARY ATTRACTORS IN NO-SCALE SUPERGRAVITY

The generalized α -attractor potentials introduced in Eqs. (39) and (40) are not just phenomenological constructions, but can be derived rather easily from fundamental supergravity theories. This connection provides both theoretical motivation and constraints on the allowed parameter space. We demonstrate here how these potentials emerge from no-scale supergravity [17,18] with specific choices of the superpotential.

We start with the no-scale Kähler potential given in Eq. (8). As discussed earlier, the parameter α determines the curvature of the Kähler manifold, with $R = 2/(3\alpha)$ [131], and recall that the scalar potential in supergravity is given by

$$V = e^{K/M_P^2} \left[K^{i\bar{j}} D_i W D_{\bar{j}} \bar{W} - \frac{3}{M_P^2} |W|^2 \right], \quad (\text{A1})$$

where $D_i W = \partial_i W + \frac{K_i}{M_P^2} W$ is the Kähler covariant derivative, and $K^{i\bar{j}}$ is the inverse Kähler metric.

In Sec. II, we described the simple derivation of the Starobinsky potential (which is identical with that in the E-model with $\alpha = 1$) from either of the superpotentials (9) or (11), which both lead to Eq. (2) once T or ϕ , respectively is fixed (stabilized) [19,21]. Similarly, the T-model potential (7) with $\alpha = 1$ can be derived using [109]

$$W = \frac{1}{4} \sqrt{\lambda} \left(M_P \phi^2 - \frac{1}{6} \frac{\phi^4}{M_P} \right), \quad (\text{A2})$$

with $T = \frac{1}{2} M_P$ fixed, or

$$W = \sqrt{\frac{3}{4}} \sqrt{\lambda} M_P \phi (2T) \left(\frac{2T/M_P - 1}{2T/M_P + 1} \right), \quad (\text{A3})$$

with $\phi = 0$ fixed.

The generalization of the T-model potentials with $V(\varphi) \propto \varphi^k$ near the minimum but still with $\alpha = 1$ are also easily expressed in terms of ϕ and T

$$W = 3^{\frac{1}{2}-\frac{k}{4}} \sqrt{\lambda} M_P^3 \left(\frac{1}{k+2} \left(\frac{\phi}{M_P} \right)^{\frac{k}{2}+1} - \frac{1}{3(k+6)} \left(\frac{\phi}{M_P} \right)^{\frac{k}{2}+3} \right). \quad (\text{A4})$$

or

$$W = \sqrt{\frac{3}{4}} \sqrt{\lambda} M_P \phi (2T) \left(\frac{2T/M_P - 1}{2T/M_P + 1} \right)^{\frac{k}{2}}, \quad (\text{A5})$$

The superpotential for the generalized E-model cannot be expressed as a simple polynomial of ϕ as in Eq. (9) for $k = 2$ or as in Eq. (A4) for the generalized T-models, but can be expressed in terms of a hypergeometric function ${}_2F_1(-1 + \frac{k}{2}, 1 + \frac{k}{2}, 2 + \frac{k}{2}, -\frac{\phi}{\sqrt{3}})$.

There are several general forms of the superpotential that lead to relatively simple forms for the scalar potential. These are useful for finding the corresponding superpotential for many of the types of models discussed here. For example, taking

$$W = A \sqrt{\lambda} M_P^3 \left(\frac{T}{M_P} - \frac{1}{2} \right) f \left(\frac{\phi}{M_P} \right), \quad (\text{A6})$$

gives

$$V = 3A^2 \lambda M_P^4 \frac{|f(\frac{\phi}{M_P})|^2}{(3 - |\phi/M_P|^2)^2}, \quad (\text{A7})$$

when one fixes $T = \frac{1}{2} M_P$, and the canonical inflaton is given by Eq. (10). Similarly,

$$W = A \sqrt{\lambda} M_P^2 \phi \left(\frac{2T}{M_P} \right) f \left(\frac{T}{M_P} \right), \quad (\text{A8})$$

gives

$$V = 4A^2 \lambda M_P^4 \frac{|T|^2}{(T + \bar{T})^2} \left| f \left(\frac{T}{M_P} \right) \right|^2, \quad (\text{A9})$$

when one fixes $\phi = 0$ and the canonical inflaton is given by Eq. (12).

In this case, the generalized E-models are very simply obtained from Eq. (A6) with

$$f \left(\frac{\phi}{M_P} \right) = \left(\frac{\phi/M_P}{\phi/M_P + \sqrt{3}} \right)^{\frac{k}{2}} (3 - \phi^2/M_P^2), \quad (\text{A10})$$

using $A^2 = 2^{k-2}$. We then fix the fields as

$$\langle T \rangle = \frac{1}{2} M_P, \quad \phi = \bar{\phi} \quad (\text{real modulus}), \quad (\text{A11})$$

and use the canonical field normalization in Eq. (10). Alternatively, the same potential can be obtained from Eq. (A8), and

$$f\left(\frac{T}{M_P}\right) = \left(1 - \frac{M_P}{2T}\right)^{\frac{k}{2}}, \quad (\text{A12})$$

with $A^2 = \frac{3}{4}$. The inflationary trajectory is now characterized by

$$\langle\phi\rangle = 0, \quad T = \bar{T} \quad (\text{real modulus}), \quad (\text{A13})$$

and we must now use the field redefinition in Eq. (12) to normalize the inflaton kinetic term.

Using Eq. (A6), the T-models are even more easily obtained by choosing

$$f\left(\frac{\phi}{M_P}\right) = \left(\frac{\phi}{M_P}\right)^{\frac{k}{2}} \left(3 - \frac{\phi^2}{M_P^2}\right), \quad (\text{A14})$$

with $A^2 = \frac{1}{4}3^{-k/2}$. The corresponding function of $f(T)$ can be read off from Eq. (A5).

When $\alpha \neq 1$, the choice of superpotential is somewhat more complicated. We can use the same form for the superpotential given in Eq. (A6) or Eq. (A8) and obtain

$$V = 3^{3\alpha-2} A^2 \lambda M_P^4 \frac{|f(\phi/M_P)|^2}{\alpha(3 - |\phi/M_P|^2)^{3\alpha-1}}, \quad (\text{A15})$$

or

$$V = 4A^2 \lambda M_P^4 \frac{|T/M_P|^2 |f(T/M_P)|^2}{\alpha(T/M_P + \bar{T}/M_P)^{3\alpha-1}}, \quad (\text{A16})$$

respectively. The generalized E-model (39) is then obtained with either

$$f(\phi/M_P) = \left(\frac{\phi/M_P}{\phi/M_P + \sqrt{3}}\right)^{\frac{k}{2}} \left(3 - \frac{\phi^2}{M_P^2}\right)^{\frac{(3\alpha-1)}{2}}, \quad (\text{A17})$$

or

$$f(T/M_P) = \left(1 - \frac{M_P}{2T}\right)^{\frac{k}{2}} \left(\frac{T}{M_P}\right)^{(3\alpha-3)/2}, \quad (\text{A18})$$

with $A^2 = 2^{k-2}3^{(3-3\alpha)}\alpha$ or $A^2 = 3 \cdot 2^{(3\alpha-5)}\alpha$ and using the canonical field normalization

$$\phi = \sqrt{3}M_P \tanh\left(\frac{\varphi}{\sqrt{6}\alpha M_P}\right), \quad (\text{A19})$$

or

$$T = \bar{T} = \frac{1}{2} e^{\sqrt{\frac{2}{3\alpha}} \frac{\varphi}{M_P}}, \quad (\text{A20})$$

to obtain the canonical inflaton.

Similarly, the generalized T-model potential (40) also follows from either (A6) or Eq. (A8) with a different choice of superpotential functions

$$f(\phi/M_P) = \left(\frac{\phi}{M_P}\right)^{\frac{k}{2}} (3 - \phi^2/M_P^2)^{\frac{(3\alpha-1)}{2}}, \quad (\text{A21})$$

or

$$f(T/M_P) = \left(\frac{2T/M_P - 1}{2T/M_P + 1}\right)^{\frac{k}{2}} \left(\frac{T}{M_P}\right)^{(3\alpha-3)/2}. \quad (\text{A22})$$

Then, using $A^2 = \frac{1}{4}3^{(3-3\alpha-\frac{k}{2})}\alpha$ or $A^2 = 3 \cdot 2^{(3\alpha-5)}\alpha$ and Eq. (A19) or Eq. (A20) we recover the potential in Eq. (40).

Finally, we show how to construct Starobinsky-like and T-models with small deformations that lead to larger values of the spectral tilt n_s , which can more easily accommodate the ACT DR6 results.

In the case of the modified Starobinsky-like models, we use

$$f(\phi/M_P) = \left(\frac{\phi}{M_P}\right)^{k/2} \left(3 - \frac{\phi^2}{M_P^2}\right)^{\frac{1}{2}(3\alpha-1-k)} \times \left(\kappa \frac{\phi}{M_P} - \sqrt{3}\right)^{k/2}, \quad (\text{A23})$$

with $A^2 = 2^{k-2}3^{3-3\alpha}\alpha$, or

$$f(T/M_P) = \frac{(-1 + 4T^2/M_P^2 - \kappa(1 - 2T/M_P)^2)^{\frac{k}{2}}}{(T/M_P)^{\frac{1}{2}(3+k-3\alpha)}}, \quad (\text{A24})$$

with $A^2 = 3 \cdot 2^{(3\alpha-5-2k)}\alpha$. Evaluating the scalar potential along the inflationary trajectory in either case leads to Eq. (47). For $\kappa = 1$, we recover Eq. (39).

For the modified T-models, we use

$$f(\phi/M_P) = (3 - \phi^2/M_P^2)^{\frac{1}{2}(3\alpha-3)} \left(\frac{\phi}{M_P}\right)^{\frac{k}{2}} \left(1 - \frac{\kappa}{3} \frac{\phi^2}{M_P^2}\right), \quad (\text{A25})$$

with $A^2 = \frac{1}{4}3^{5-3\alpha-\frac{k}{2}}\alpha$, or

$$f(T/M_P) = \left(\frac{2T/M_P - 1}{2T/M_P + 1}\right)^{\frac{k}{2}} \left(\frac{T}{M_P}\right)^{(3\alpha-5)/2} \times ((1 + 2T/M_P)^2 - \kappa(1 - 2T/M_P)^2), \quad (\text{A26})$$

with $A^2 = 3 \cdot 2^{3\alpha-11}\alpha$ which, combined with Eq. (A19) or Eq. (A20), leads to Eq. (48) and reduces to Eq. (40) when $\kappa = 1$.

APPENDIX B: ANALYTICAL APPROXIMATIONS FOR INFLATIONARY OBSERVABLES

While the results presented in the main text are obtained through numerical integration, we provide here analytical approximations that offer insight into the parameter dependencies and serve as useful benchmarks for numerical calculations.

Solving the condition (19) for the α -attractor potentials yields the following field values at the end of inflation. For E-models

$$\frac{\varphi_{\text{end}}}{M_P} \simeq \sqrt{\frac{3\alpha}{2}} \ln \left[\frac{2(6\alpha + 3\sqrt{3\alpha} - 2)}{12\alpha - 1} \right], \quad (\text{B1})$$

and for T-models

$$\begin{aligned} \frac{\varphi_{\text{end}}}{M_P} \simeq & \sqrt{\frac{3\alpha}{2}} \ln \left[\frac{4 - 6\sqrt{\alpha(5+4\alpha)}}{1 - 12\alpha} \right] \\ & + \sqrt{\frac{75}{5 + 68\alpha + 16\sqrt{\alpha(5+4\alpha)}}}. \end{aligned} \quad (\text{B2})$$

For the canonical case $\alpha = 1$, these expressions reduce to

$$\alpha\text{-Starobinsky: } \varphi_{\text{end}}/M_P = 0.63, \quad (\text{B3})$$

$$\text{T-model: } \varphi_{\text{end}}/M_P = 0.89, \quad (\text{B4})$$

recovering the values given in Eqs. (24) and (26).

The field value φ_* when the pivot scale exits the horizon is determined by integrating the number of e -folds from Eq. (18). The analytical solutions for the α -Starobinsky model are

$$\begin{aligned} \frac{\varphi_*}{M_P} \simeq & \sqrt{\frac{3\alpha}{2}} \left[1 + \frac{3\alpha}{4N_* - 3\alpha} \right] \\ & \times \ln \left(\frac{4N_*}{3\alpha} + e^{\sqrt{\frac{2}{3}} \frac{\varphi_{\text{end}}}{M_P}} - \sqrt{\frac{2}{3}} \frac{\varphi_{\text{end}}}{M_P} \right), \end{aligned} \quad (\text{B5})$$

and for the T-model

$$\frac{\varphi_*}{M_P} \simeq \sqrt{\frac{3\alpha}{2}} \cosh^{-1} \left[\frac{4N_*}{3\alpha} + \cosh \left(\sqrt{\frac{2}{3}} \frac{\varphi_{\text{end}}}{M_P} \right) \right], \quad (\text{B6})$$

which reduce to Eqs. (23) and (25) when $\alpha = 1$. We have compared these analytical approximations with exact numerical results across the parameter ranges relevant for CMB constraints. The relative errors are summarized in Table III.

The approximations for φ_* are particularly accurate, with errors below 1% for $\alpha \lesssim 1$. The larger errors for φ_{end} at large α values reflect the breakdown of the slow-roll approximation near the end of inflation. Nevertheless,

TABLE III. Maximum relative errors in the analytical approximations for $40 < N_* < 60$.

Model	$\alpha = 0.1$	$\alpha = 1$	$\alpha = 10$
	φ_{end}/M_P		
α -Starobinsky	2%	2%	4%
T-model	3%	5%	5%
	φ_*/M_P		
α -Starobinsky	0.3%	0.3%	3%
T-model	0.4%	0.5%	0.7%

these analytical expressions provide reliable estimates for cosmological observables and serve as efficient starting points for numerical calculations.

The postinflationary evolution significantly affects the relationship between the number of e -folds N_* and the inflationary parameters. For $k = 2$, simplification of Eq. (27) for N_* as a function of the reheating temperature is possible using the relation [84]

$$\frac{1 - 3w_{\text{int}}}{12(1 + w_{\text{int}})} \ln \left(\frac{\rho_{\text{rad}}}{\rho_{\text{end}}} \right) \simeq \frac{1}{6} \ln \left(\frac{\Gamma_{\varphi}}{H_{\text{end}}} \right). \quad (\text{B7})$$

At the pivot scale $k_* = 0.05 \text{ Mpc}^{-1}$ we can therefore write

$$\begin{aligned} N_* \simeq & 61.41 + \frac{1}{6} \ln \lambda + \frac{1}{2} \ln \left(\frac{V_*}{\lambda M_P^4} \right) \\ & - \frac{1}{3} \ln \left(\frac{V_{\text{end}}}{\lambda M_P^4} \right) + \frac{1}{3} \ln \left(\frac{T_{\text{RH}}}{M_P} \right), \end{aligned} \quad (\text{B8})$$

where the dependence on α can be obtained by substitution of (B1) and (B5) into the α -Starobinsky potential (6), or by substitution of (B2) and (B6) into the T-model potential (7). The dependence on the coupling λ is explicitly separated, and requires the substitution (38). For $\alpha = 1$ this simplifies to

$$N_* \simeq 59.55 - \frac{1}{3} \ln N_* + \frac{1}{3} \ln \left(\frac{T_{\text{RH}}}{M_P} \right), \quad (\text{B9})$$

for α -Starobinsky models, and for T-models

$$N_* \simeq 59.67 - \frac{1}{3} \ln N_* + \frac{1}{3} \ln \left(\frac{T_{\text{RH}}}{M_P} \right). \quad (\text{B10})$$

Comparison with full numerical calculations shows that these analytical approximations achieve remarkable accuracy. We find a maximal error of 0.2% for α -Starobinsky models and a maximal error of 0.1% for T-models across the parameter range shown in Fig. 2. These analytical results provide important insight into how reheating dynamics affects inflationary predictions, demonstrating that uncertainties in the reheating temperature translate into relatively modest shifts in N_* due to the logarithmic

dependence. For comprehensive analyses of reheating in α -attractor models, see [132,133].

For $k > 2$ Eq. (19) can be solved to obtain approximate expressions for the inflaton value at the end of inflation. For the E-models this yields

$$\frac{\varphi_{\text{end}}}{M_P} \simeq \sqrt{\frac{3\alpha}{2}} \ln \left[\frac{2(6\alpha + \sqrt{3\alpha(4k^2 - 4k + 1)} - k)}{12\alpha - 1} \right]. \quad (\text{B11})$$

while for T-models

$$\begin{aligned} \frac{\varphi_{\text{end}}}{M_P} \simeq & \sqrt{\frac{3\alpha}{2}} \ln \left[\frac{2k(1 - 2\sqrt{3\alpha})}{1 - 12\alpha} \right] \\ & + \sqrt{\frac{(1 + 12\alpha - 4\sqrt{3\alpha})(12\alpha + 4k^2 - 1)}{(12\alpha - 1)^2}}. \end{aligned} \quad (\text{B12})$$

Similarly, the corresponding expressions for the field value at the horizon exit of the pivot scale are given by

$$\begin{aligned} \frac{\varphi_*}{M_P} \simeq & \sqrt{\frac{3\alpha}{2}} \left[1 + \frac{3\alpha}{2kN_* - 3\alpha} \right] \\ & \times \ln \left(\frac{2kN_*}{3\alpha} + e^{\sqrt{\frac{2}{3}} \frac{\varphi_{\text{end}}}{M_P}} - \sqrt{\frac{2}{3}} \frac{\varphi_{\text{end}}}{M_P} \right), \end{aligned} \quad (\text{B13})$$

for the E-models, and

$$\frac{\varphi_*}{M_P} \simeq \sqrt{\frac{3\alpha}{2}} \cosh^{-1} \left[\frac{2kN_*}{3\alpha} + \cosh \left(\sqrt{\frac{2}{3}} \frac{\varphi_{\text{end}}}{M_P} \right) \right], \quad (\text{B14})$$

for the T-models. With these expressions at hand, the equivalent of Eq. (B8) takes the form

$$\begin{aligned} N_* \simeq & 61.49 + \ln \left[\left(\frac{3}{2} \right)^{\frac{k+2}{6k}} \left(\frac{\pi^2}{30} \right)^{\frac{k+4}{12k}} \right] + \frac{k-1}{3k} \ln \lambda \\ & + \frac{1}{2} \ln \left(\frac{V_*}{\lambda M_P^4} \right) - \frac{k+2}{6k} \ln \left(\frac{V_{\text{end}}}{\lambda M_P^4} \right) \\ & - \frac{k-4}{3k} \ln \left(\frac{\max[T_{\text{RH}}, T_{\text{frag}}]}{M_P} \right) - \frac{k-2}{6k} \ln g_{\text{RH}}, \end{aligned} \quad (\text{B15})$$

where T_{frag} denotes the temperature of the radiation at the moment of inflaton fragmentation originated from the self-interaction of the inflaton (see the corresponding discussion in Sec. VII B). We note that for $k = 4$ and $\alpha = 1$ this gives $N_* \simeq 55.7$ (E-model) and $N_* \simeq 55.8$ (T-model) for the MSSM degrees of freedom, independent of the details of reheating.

APPENDIX C: NUMERICAL COMPUTATION OF INFLATIONARY OBSERVABLES

While the slow-roll approximations presented in Sec. IV provide valuable analytical insights, precise comparison

with CMB data requires numerical integration of the perturbation equations. This is particularly important for models near the boundary of observational constraints, where percent-level accuracy in n_s can determine viability.

We compute the primordial scalar power spectrum by solving the equation of motion for gauge-invariant curvature perturbations. In the uniform-density gauge, the Mukhanov-Sasaki variable is defined as

$$Q_k = \delta\varphi_k + \frac{\dot{\varphi}}{H} \Psi_k, \quad (\text{C1})$$

where $\delta\varphi_k$ and Ψ_k are the Fourier modes of the inflaton and metric perturbations, respectively. The evolution equation for Q_k follows from the perturbed Einstein equations [134,135]

$$\ddot{Q}_k + 3H\dot{Q}_k + \left[\frac{k^2}{a^2} + m_{\text{eff}}^2 \right] Q_k = 0, \quad (\text{C2})$$

where the effective mass term is

$$m_{\text{eff}}^2 = V_{\varphi\varphi} - \frac{\dot{\varphi}^4}{2H^2} + \frac{2\dot{\varphi}V_{\varphi}}{H} + 3\dot{\varphi}^2. \quad (\text{C3})$$

We impose Bunch-Davies initial conditions in the sub-horizon limit

$$Q_k = \frac{1}{a\sqrt{2k}} e^{-ik\tau} \quad \text{for } k \gg aH, \quad (\text{C4})$$

where $\tau = \int dt/a$ is the conformal time. The comoving curvature perturbation is related to the Mukhanov-Sasaki variable by

$$\mathcal{R}_k = \frac{H}{\dot{\varphi}} Q_k. \quad (\text{C5})$$

The dimensionless power spectrum, defined through

$$\langle \mathcal{R}_k \mathcal{R}_{k'}^* \rangle = \frac{2\pi^2}{k^3} \mathcal{P}_{\mathcal{R}}(k) \delta^{(3)}(\mathbf{k} - \mathbf{k}'), \quad (\text{C6})$$

is evaluated numerically at horizon crossing ($k = aH$) for each mode. The spectral index is then computed from its definition at the end of inflation

$$n_s - 1 = \left. \frac{d \ln \mathcal{P}_{\mathcal{R}}}{d \ln k} \right|_{k=k_*}. \quad (\text{C7})$$

Analogously, we solve numerically for the tensor perturbations, using the standard transverse, traceless perturbation. The equation of motion satisfied by the tensor mode functions is given by

$$\ddot{h}_{k,\gamma} + 3H\dot{h}_{k,\gamma} + \frac{k^2}{a^2}h_{k,\gamma} = 0. \quad (\text{C8})$$

where $\gamma = +, \times$ denotes the two polarization states. The corresponding power spectrum is defined as

$$\sum_{\gamma=+,\times} \langle h_{k,\gamma} h_{k',\gamma}^\dagger \rangle = \frac{2\pi^2}{k^3} \mathcal{P}_T(k) \delta^{(3)}(\mathbf{k} - \mathbf{k}'), \quad (\text{C9})$$

with the tensor-to-scalar ratio computed then as follows,

$$r = \left. \frac{\mathcal{P}_T}{\mathcal{P}_R} \right|_{k=k_*}. \quad (\text{C10})$$

Our numerical results reveal appreciable corrections to the slow-roll predictions. Figure 9 illustrates the relationship between n_s and N_* computed using three methods

- (1) *Full numerical integration* (solid line): Exact solution of Eq. (C2);
- (2) *Potential slow-roll* (dashed line): Using parameters from Eq. (17);
- (3) *Hubble slow-roll* (dotted line): Using the Hubble flow parameters:

$$\varepsilon_H = -\frac{\dot{H}}{H^2} = \frac{\dot{\varphi}^2}{2M_P^2 H^2}, \quad (\text{C11})$$

$$\eta_H = 2\varepsilon_H - \frac{\dot{\varepsilon}_H}{2\varepsilon_H H}. \quad (\text{C12})$$

The potential slow-roll approximation exhibits systematic errors of $\Delta N_* \gtrsim 1$ when matching to a given value of n_s . This discrepancy arises from violations of slow-roll near the end of inflation, higher-order gradient corrections in the

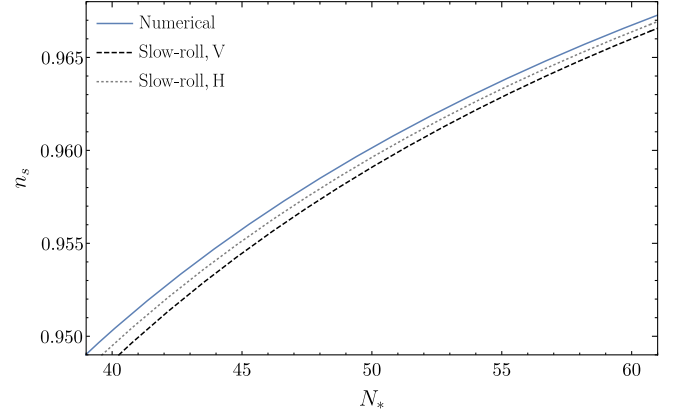


FIG. 9. The spectral tilt, n_s , as a function of the number of e -folds N_* for the α -Starobinsky model with $\alpha = 1$. The solid blue line shows the exact numerical solution of the Mukhanov-Sasaki equation (C2). The dotted gray line represents the slow-roll approximation using Hubble flow parameters (C11), while the dashed black line uses potential slow-roll parameters (17). The shaded region indicates the 68% CL constraint from *Planck* 2018 data.

perturbation equations, and the evolution of slow-roll parameters during horizon crossing. The Hubble slow-roll approximation provides improved accuracy as it better captures the instantaneous dynamics. For precision cosmology, however, full numerical integration remains essential, particularly when confronting models with tight observational constraints such as those from ACT DR6.

The accuracy of different approximation schemes becomes apparent when comparing predictions for the spectral tilt as a function of e -fold number. Figure 9 demonstrates these differences for the canonical Starobinsky model ($\alpha = 1$).

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